

The Partition Principle does not imply Choice

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Abstract

We prove that the Partition Principle does not imply the Axiom of Choice: if ZF is consistent, then so is ZF with the Partition Principle, Choice for ordinal-indexed families, and the negation of the Axiom of Choice. Separately, over every countable transitive model of ZFC, we construct a transitive symmetric model of this theory with the same ordinals and no new countable sequences of ground elements.

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1 Introduction

The *Partition Principle* asserts that a surjection between two sets implies the existence of an injection in the reverse direction:

$$\text{PP} : \quad \forall X \forall Y [(\exists f : X \twoheadrightarrow Y) \implies (\exists j : Y \hookrightarrow X)].$$

Equivalently, every partition of a set admits an injection into that set. The Axiom of Choice implies this principle by choosing one element of each fiber. The converse is less direct: the injection supplied by PP need not take a part to one of its members. In particular, it need not satisfy $f \circ j = \text{id}_Y$. The problem is whether this cardinal comparison alone can recover choice for arbitrary families.

Write AC_{WO} for Choice for wellorderable families: for every ordinal η and every family $(X_\xi)_{\xi < \eta}$ of nonempty sets, there is a function c with $c(\xi) \in X_\xi$ for all $\xi < \eta$. Our principal result is the following ordinary relative-consistency statement.

Theorem 1.1. *If ZF is consistent, then so is $\text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \neg \text{AC}$.*

The construction also has a separate transitive-model form.

Theorem 1.2. *Let V be a countable transitive model of ZFC. There are a set forcing $\mathbb{Q} \in V$, a V -generic filter G , and a transitive symmetric submodel W of $V[G]$ such that*

$$V \subseteq W \subseteq V[G], \quad \text{Ord}^W = \text{Ord}^V, \quad W \models \text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \neg \text{AC}.$$

Every countable sequence of elements of V that belongs to W already belongs to V .

Here ZF is classical first-order set theory with pure sets, Foundation, and the full Separation and Replacement schemata. Theorem 1.1 gives a negative answer to the Partition Principle versus Choice problem under ordinary syntactic consistency. It permits externally ill-founded models, whereas Theorem 1.2 starts with a transitive ground and preserves transitivity. In either conclusion the reverse injections and the family without a choice function belong to the same model. Section 8 gives the two deductions separately. The consequences for the dual Cantor–Schröder–Bernstein principle and the Weak Partition Principle are developed in Remark 7.3.

1.1 History and related methods

The distinction between comparing the sizes of a partition and its underlying set, and actually choosing representatives, is longstanding. Da Silva traces the Partition Principle to Beppo Levi in 1902 [4, Section 2, p. 787]. Banaschewski and Moore studied its relation to the dual Cantor–Bernstein theorem and the surrounding cardinal-comparison principles [1]. These questions concern what remains of cardinal arithmetic when sets need not be wellorderable. For example, Blass and Kulshreshtha distinguish several forms of cardinal well-foundedness and show that they become equivalent under PP; their August 2026 version still lists $\text{PP} \Rightarrow \text{AC}$ as open [3, Introduction and Section 9]. Gilson’s September 2026 corrections withdraw the model-construction claims of three earlier approaches [8, 7, 6].

There is already substantial choice in PP. The implication $\text{PP} \Rightarrow \text{AC}_{\text{WO}}$ is classical: Ryan-Smith records the argument attributed to Pincus by Pelc [15, Proposition 3.17, proof and footnote 4]. Thus AC_{WO} in our main statements is not an additional strength beyond PP. Its separate appearance records the order of the construction: we establish ordinal-indexed choice first and then use it to assemble reverse injections for arbitrary sets.

The difficulty of preserving a principle quantified over all sets can be reduced to a fixed set of parameters. Blass introduced small violations of choice [2]. For a nonempty set A , $\text{SVC}(A)$ says that every nonempty set is a surjective image of $A \times \eta$ for some ordinal η . The injective version, $\text{SVC}^+(A)$, says that every set embeds into some $A \times \eta$. See also [16, Definition 3.13 and Theorem 3.14]. These presentations let a construction on one set control objects of arbitrary rank, but their different directions cannot be interchanged without proof.

Ryan-Smith makes the local-to-global step precise. Under $\text{SVC}^+(A)$, full PP is equivalent to AC_{WO} together with reverse injections for surjections between subsets of A [15, Proposition 3.17]. Reverse injections for surjective images of A are a different local assertion: even with AC_{WO} and $\text{SVC}^+(A)$, they need not imply full PP [15, Proposition 3.23]. Our construction instead makes every nonwellorderable image of A bijective to A . Lemma 7.1 uses this stronger classification, the surjective presentation, and ordinal-indexed choice to carry out the required all-set reduction. Its slicing argument follows the method of [15, Propositions 3.17 and 3.20], rather than identifying the two local principles.

The ambient method is a symmetric submodel of a set-forcing extension: hereditarily symmetric names for a normal filter of subgroups give a model of full ZF [11, Section 2.1]. A more specific antecedent is Holy and Schilhan's product of Cohen forcing with a forcing of ground-model hereditarily almost-disjoint towers [9, Section 3]. Their construction yields models in which every set can be linearly ordered and higher forms of Dependent Choice hold, although full Choice fails. They develop the hereditary almost-disjointness method from Pincus's work on adding Dependent Choice [14]. Their restriction and support-intersection arguments [9, Lemmas 10 and 15] are antecedents of the locality arguments here. Our diagrams have a different geometry: the monoid must support both exact quotient multiplicities and the coherent assignment of bijections, not just the descent of supports.

1.2 Proof strategy

We construct a set A of distinct generic subsets of the ground-model ω_1 . Symmetry prevents A from being wellordered. Orbit maps of names show that every nonempty set is a surjective image of $A \times \eta$ for some ordinal η , with no restriction on its rank. The forcing also allows one wellorderable part of A to meet every member of any given ordinal-indexed family of nonempty subsets of A . Together with the preceding presentation of arbitrary sets, this yields AC_{WO} .

The central additional property is

$$\text{every nonwellorderable surjective image of } A \text{ is in bijection with } A. \quad (1.1)$$

To pass from this property to full PP, partition the target of an arbitrary surjection into ordinal-indexed disjoint nonempty pieces and use disjoint source pieces mapping onto them. Each resulting piece is an image of A , though the source pieces need not cover the original domain. A wellorderable target piece admits a reverse injection by AC_{WO} . Otherwise both corresponding pieces are in bijection with A by (1.1). A further application of AC_{WO} chooses the piecewise injections, whose union is the required injection. The empty target uses the empty injection.

The difficult part is to construct the bijections in (1.1) with enough coherence to belong to the symmetric model. Fix one equivalence relation on A whose quotient is not wellorderable. Counting its classes on each support will produce many possible local bijections, but arbitrary choices of them need not agree on overlapping supports or respect coordinate changes. The proof solves these two compatibility problems in three stages.

1. **A geometry of diagrams.** We build a left-cancellative monoid T . A diagram is an injective map $r : T \rightarrow I$, where I is the set of label indices. Its subdiagram at $n \in T$ is the map $r_n(x) = r(nx)$. The diagram forcing makes larger diagrams stronger conditions. Finite amalgamations join independently placed diagrams. A second extension operation places many copies with prescribed patterns of intersection. Together these operations give the forcing its compatibility and support-descent properties. Height and cost functions reduce the availability patterns introduced next to countable cofinal paths. A rearrangement of the same monoid construction will later bound the symmetries that survive along such a path.
2. **Exact availability fibers.** A diagram supports names whose definitions are fixed by all permutations of labels outside its range. For a quotient of A , we record the quotient elements that can be named at each diagram. An element's availability pattern records the subdiagrams at which it can still be named. The labels themselves have analogous patterns, determined by which subdiagrams contain their indices. The copying and support-descent lemmas show that, for each exact pattern, the available labels and quotient elements have the same cardinality. Restricting to a subdiagram in that pattern identifies the corresponding fibers canonically. These identifications group the fibers into components and are already compatible; the choices of bijections between label and quotient fibers are not.
3. **Covariant choices on each component.** Each component has a countable cofinal chain of nested diagrams. Codes along this chain give a preliminary bijection compatible with every permutation fixing a sufficiently late diagram pointwise. Modulo these permutations, the remaining symmetries form a countable group. Countably many bounded initial segments of Cohen subsets distinguish its elements and select among the transported preliminary bijections. Finally, transport between components gives assignments that agree under restriction and respect all required permutations. Their union is one hereditarily symmetric graph for the fixed quotient.

The countable description of each availability pattern, the reduction of its symmetries to a countable group, and the final equivariant selection are the main technical ingredients. The selection is performed between moved Cohen generics: it does not require permutations to act as automorphisms of a single fixed forcing extension.

Section 2 constructs T and proves its local rearrangement property. Section 3 develops the forcing, locality, and the basic properties of A . The quotient construction follows in three stages: Section 4 computes exact availability fibers, Section 5 organizes them into components and controls their symmetries, and Section 6 constructs the coherent bijections. Section 7 derives full PP and identifies a family without a choice function. Section 8 gives the ordinary consistency deduction and the separate transitive-ground result.

2 A monoid of diagrams

The forcing construction will compare a diagram with its translated subdiagrams. We first construct the monoid that indexes these translations. Its two extension operations serve different purposes: one realizes finitely many prescribed overlaps, and the other inserts many new positions with a prescribed set of old divisors. A further property of the construction will control symmetries along countable systems of subdiagrams.

All constructions in this section take place in a universe satisfying ZFC. No inaccessible cardinal is used. Fix an infinite cardinal $\mu \geq \aleph_1$ such that $\mu^{\aleph_0} = \mu$.

2.1 The required algebra and the construction theorem

Let M be a monoid with identity 1. Write $x \sqsubseteq y$ if $xz = y$ for some z . We use only left cancellation. A monoid is *conical* if $xy = 1$ implies $x = y = 1$. Left cancellation and conicality make $\sqsubseteq$ a partial order and make the right factor in a comparison unique. Write $x \sqsubset y$ for $x \sqsubseteq y$ and $x \neq y$, and $\downarrow a = \{x : x \sqsubseteq a\}$. A *conditional join* is an element $x \vee y$ satisfying

$$xM \cap yM = (x \vee y)M$$

whenever the intersection is nonempty. Thus a join is required exactly when the two elements have a common right multiple. An *ideal* in $\downarrow a$ will mean a nonempty downward closed subset closed under finite nonempty joins. These joins exist because a is an upper bound.

An inclusion $B \subseteq M$ is *conservative* if it preserves and reflects comparisons, preserves their right factors, preserves existing joins, and preserves the absence of a common right multiple. In particular, if $x, y \in B$ and $xz = y$ in M , then $z \in B$ and the same equation holds in B . Mere injectivity is not this property.

We shall also keep real-valued height and cost functions ρ, ϵ on M , writing $\epsilon_x = \epsilon(x)$, and satisfying

$$\begin{aligned} 0 < \rho(x) < 1, \quad 0 < \epsilon_x \leq 1, \\ \rho(y) - \rho(x) > \sup\{\epsilon_d : d \sqsubseteq y, d \not\sqsubseteq x\} \quad (x \sqsubset y). \end{aligned} \tag{2.1}$$

An empty supremum of costs is understood to be zero.

Lemma 2.1. *If a monoid has conditional joins and satisfies Equation (2.1), every ideal in $\downarrow a$ has a nondecreasing cofinal sequence of length ω , with repetitions allowed. The same conclusion holds in any ZFC extension in which the monoid, order, heights and costs retain their meanings, for ideals belonging to that extension.*

Proof. For an ideal U , choose elements whose heights approach $\sup \rho[U]$, and replace their successive initial segments by their finite joins. This gives a nondecreasing sequence (n_i) in U with heights approaching the same supremum. If $d \in U$ divides none of the n_i , then

$$\rho(n_i \vee d) - \rho(n_i) > \epsilon_d$$

for every i , contradicting convergence to the supremum. Prepending 1 makes $n_0 = 1$. The proof applies in the stated extension as well; its choices are made in that ZFC universe. $\square$

Here is the finite overlap condition used below. Take finitely many copies $\{i\} \times M$. For each pair of distinct copies prescribe either no overlap or elements $a_{ij}, a_{ji} \in M$ and the identifications

$$(i, a_{ij}x) \sim (j, a_{ji}x) \quad (x \in M).$$

We call the prescription *valid* if these pair identifications, together with equality, already form an equivalence relation and do not identify distinct elements of any one copy. In particular, transitive closure is not allowed to create extra overlaps.

Theorem 2.2. *There is a monoid T of cardinality μ , with a distinguished $e \neq 1$, having the following properties.*

1. *T is left cancellative and conical, has conditional joins, and has heights and costs satisfying Equation (2.1).*

2. Every valid finite overlap prescription on copies of T is realized by left translations $x \mapsto l_i x$ in T , with exactly the prescribed identifications.

3. For $a, b \in T$ with $b \neq 1$, and every ideal $U \subseteq \downarrow a$, there are μ distinct elements m with

$$\begin{aligned} mb = ae, \quad m \not\sqsubseteq a, \quad \downarrow m \cap \downarrow a = U, \\ n \vee m = ae \quad (n \sqsubseteq a, n \notin U). \end{aligned} \tag{2.2}$$

The cones of distinct elements in this family intersect exactly in aeT . Moreover $|T \setminus eT| = \mu$.

4. For every countable subset of T there is a countable submonoid T_0 containing it and a continuous conservative tower starting at T_0 and ending at T . Each successor step has one of the following properties:

(a) no new element divides an old element;

(b) every new element w with an old right multiple has a least old right multiple $g(w)$, and

$$d \text{ old}, \quad g(w) \sqsubseteq wd \implies wd \text{ old}. \tag{2.3}$$

Here “old” means belonging to the base of that particular successor step, and the comparisons may be tested in the final monoid.

For diagrams indexed by T , the finite-merger clause supplies a common larger diagram for finitely many copies with compatible overlaps. The probe clause supplies μ copies with one prescribed intersection pattern: for $n \sqsubseteq a$, the cone nT contains mT exactly when $n \in U$, and otherwise meets it in aeT . The margin makes every such ideal countably cofinal. The countable-base clause serves a different purpose: it will allow suitably extendible positions below a countable path to be moved into a countable submonoid. We formulate that descent precisely at the end of the section and use it to bound the residual symmetries of diagrams.

We prove the theorem through two extension lemmas. They preserve more than the order: keeping right factors and nonjoins fixed is what permits the operations to be rearranged over a countable submonoid.

2.2 The extension that inserts a probe

Suppose M has all the algebraic and margin properties above. Fix $a, b, e \in M$, with $b, e \neq 1$, and a nondecreasing path

$$1 = n_0 \sqsubseteq n_1 \sqsubseteq \cdots \sqsubseteq a, \quad n_i d_i = n_{i+1}, \quad n_i c_i = a. \tag{2.4}$$

Left cancellation gives $d_i c_{i+1} = c_i$. Form the monoid presentation over M with new generators p_i and relations

$$p_i = d_i p_{i+1}, \quad p_i b = c_i e \quad (i < \omega). \tag{2.5}$$

Put $m = p_0$. Since $n_0 = 1$ and $n_i d_i = n_{i+1}$, the first relations give $n_i p_i = m$ for every i . Thus every point of the path divides the new probe. The second relations make both factorizations in Figure 1 end at ae :

$$n_i p_i b = n_i c_i e = ae, \quad mb = ae.$$

These identities explain the presentation, but do not yet establish its exact divisor profile. We must prove that no old divisor of a outside the path’s downward closure is inserted below m , that m does not divide a , and that every omitted divisor meets m first at ae . The next lemma proves these claims even for new divisors of a introduced by the same extension.

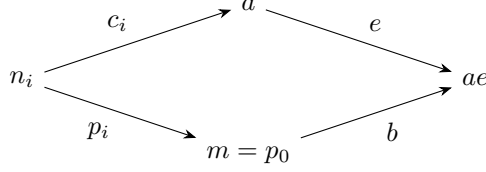


Figure 1: The relations defining the probe give two factorizations of the prescribed common multiple ae . Each arrow means right multiplication by its label, not a covering relation. This is not a Hasse diagram: path points may coincide with a , and a may divide m . The exact-profile assertion is that a divisor of a omitted from the path's downward closure has join ae with m .

Lemma 2.3. *The presented extension M' is left cancellative and conical, has conditional joins, and conservatively contains M . Every new element with an old right multiple has a least old right multiple, and Equation (2.3) holds at this extension. For $m = p_0$,*

$$\begin{aligned}
 mb = ae, \quad m \not\sqsubseteq a, \quad \downarrow m \cap \downarrow a = \bigcup_i \downarrow n_i, \\
 x \vee m = ae \quad (x \sqsubseteq a, x \not\sqsubseteq m),
 \end{aligned} \tag{2.6}$$

where the downsets and joins in this display are computed in M' .

Proof. Normal forms. Let S be the direct limit of the sets M under the maps $x \mapsto xd_i$. Write $[x, i]$ for the image of x at stage i . Equality in this direct limit means equality at a finite common stage. The transition maps need not be injective: no right cancellation has been assumed. Nevertheless the left M -action on S is injective for each multiplier. Indeed, equality after multiplying two letters by an old u becomes $uX = uY$ at a common stage, where left cancellation gives $X = Y$.

There are well-defined equivariant maps

$$k([x, i]) = xc_i, \quad h(s) = k(s)e.$$

The identity $d_i c_{i+1} = c_i$ proves well-definedness. Before imposing the second relation in Equation (2.5), the normal forms are strings

$$s_1 \cdots s_j y \quad (s_i \in S, y \in M),$$

including $j = 0$. The letter $[x, i]$ represents xp_i . Old factors between new generators are absorbed into the following letter, and the transition relations are precisely equality in S . Multiplication concatenates strings, using the left action to absorb the terminal old factor of the first string.

The remaining reductions are

$$s(bt) \mapsto h(s)t, \tag{2.7}$$

where t is a letter of S or a terminal old element. The quotient t is unique because the left b -action is injective. Each reduction removes one new letter. The only overlapping reductions occur in $s(bt)(bu)$, with the final factor possibly terminal. Reducing in either order gives the same result by

$$h(bt) = bh(t), \quad h(h(s)t) = h(s)h(t).$$

Disjoint reductions commute by associativity of the action. By Newman's lemma [13] (see [5, Lemma 5.13]), these local diamonds and termination give unique irreducibles: besides the old

elements, they are strings whose first letter is arbitrary and whose later letters and terminal factor are not left-divisible by b .

Old left multipliers act injectively on these forms. Left multiplication by a letter s either prepends s , when the input is not a b -multiple, or sends bZ to $h(s)Z$. Both cases are injective. Their ranges are disjoint, because

$$s \notin h(s)S. \quad (2.8)$$

To see this, $s \in xS$ implies $x \sqsubseteq k(s)$, whereas $k(s)e \not\sqsubseteq k(s)$ by conicality and left cancellation. Thus every left multiplication is injective. A new word can become old only by consuming its first letter, at which point it is a multiple of $h(s) \neq 1$. It cannot equal 1, proving conicality.

Old cones and the first-letter blocks. For $s \in S$, let B_s consist of all irreducibles beginning with s . For old x the normal forms give

$$xM' = xM \cup \bigcup_{s \in xS} B_s. \quad (2.9)$$

At a common stage in the direct limit, $xS \cap yS$ is $(x \vee y)S$ if the old join exists, and is empty if the old cones are disjoint. Thus old comparisons, their right factors, joins and nonjoins are unchanged. Notice that this conclusion has not used a join theorem for arbitrary new words.

Exits and gates. Fix an irreducible $w = sY$. Right multiplication either leaves its first letter intact or eventually consumes that letter. We prove the following precise alternative: the set $wM' \setminus B_s$ is empty, or it is one cone $g(w)M'$ whose root $g(w)$ is old. Thus the gate will describe every exit from the fixed-head block, not only the old multiples of w .

The first letter is consumed exactly when the suffix becomes a b -multiple, so first calculate $YM' \cap bM'$. If Y is old, Equation (2.9) gives the old join $Y \vee b$, or an empty intersection. If Y is new with head t , irreducibility gives $t \notin bS$. No member of B_t is then a b -multiple. By induction on the number of letters in the suffix, all possible exits of Y form the old-rooted cone $g(Y)M'$, if there are any. Its intersection with bM' is calculated by the old join $g(Y) \vee b$.

In either case the suffix intersection is empty or is zM' for an old $z = bu$ with old u . The factor u is old by conservation of old comparisons. In the nonempty case, multiplication by s and the reduction $sbu = h(s)u$ give

$$wM' \setminus B_s = g(w)M', \quad g(w) = h(s)u. \quad (2.10)$$

This cone cannot return to B_s : every new head in it belongs to $h(s)S$, whereas $s \notin h(s)S$ by Equation (2.8). This proves the exit alternative. Every old multiple of w lies outside B_s , so $g(w)$ is in particular its least old multiple.

We record also the right factor $c(w)$ with $wc(w) = g(w)$. If Y is old, it is the old quotient of z by Y . If Y is new and $Yc(Y) = g(Y)$, it is

$$c(w) = c(Y)v, \quad g(Y)v = z. \quad (2.11)$$

All quotients in this recursion are old except the already constructed $c(Y)$. This explicit calculation will ensure conservation of right factors when the same presentation is simulated over a larger base.

Append an old d to w . If the product leaves B_s , its first letter has been consumed; that reduction can occur only after all intervening letters have been consumed. The result is then entirely old. Since the gate cone is disjoint from B_s , the hypothesis $g(w) \sqsubseteq wd$ implies just such an exit. This proves Equation (2.3).

All conditional joins. For two forms with the same head, cancel that head and intersect their suffix cones, using induction on the total number of new letters. For old x and new $w \in B_s$, either

$s \in xS$, in which case $x \sqsubseteq w$, or every common multiple must leave B_s . The latter intersection is computed through $g(w)$ and an old join. For $w \in B_s$ and $v \in B_t$ with $s \neq t$, an intersection inside B_s requires $g(v)$ to divide a member of B_s . By Equation (2.9) it then divides all of B_s , so $v \sqsubseteq w$. The symmetric case is the same. Otherwise a common multiple must leave both blocks, and the intersection is computed by the old join of their gates. Missing gates or missing old joins give an empty intersection. This proves conditional joins in every case.

The prescribed divisor profile. For any letter s , the old divisors of every element of B_s form

$$P_s = \{x \in M : s \in xS\} \sqsubseteq \downarrow k(s). \quad (2.12)$$

This is an ideal: common-stage representatives and old joins prove join closure. For the head $[1, 0]$ of $m = p_0$, it is precisely $\bigcup_i \downarrow_M n_i$, and the gate is ae . The defining relation gives $mb = ae$. Since $ae \not\sqsubseteq a$, no member of $B_{[1,0]}$ divides a ; in particular $m \not\sqsubseteq a$.

Now let x be a new divisor of a . It is outside that block and has an old gate $g(x) \sqsubseteq a$. If $x \sqsubseteq m$, the comparison leaves its block, so $g(x) \sqsubseteq m$, and the old profile puts $g(x)$ below some n_i . Conversely $x \sqsubseteq n_i$ implies $x \sqsubseteq m$. If $x \not\sqsubseteq m$, its gate also fails to divide m . The possibility $m \sqsubseteq x$ is excluded by $x \sqsubseteq a$. The gate calculation consequently gives $x \vee m = g(x) \vee m = ae$. The corresponding assertions for old x follow directly from its block membership. This proves Equation (2.6), including new divisors of a . $\square$

2.3 Extending the real heights

The gate description also controls the costs of all newly introduced divisors. This is the reason for keeping a cost as well as a height: pointwise strict inequalities alone would not control suprema at a limit stage.

Lemma 2.4. *The height and cost functions on M extend to the probe extension M' so that Equation (2.1) holds. For any old comparison, its missed-cost supremum is unchanged.*

Proof. Use P_s from Equation (2.12), and put $A_s = \sup \rho[P_s]$. We place the whole block B_s in a short height interval above its old divisors and below $\rho(h(s))$. Here $h(s)$ is the gate of the bare letter s ; the gate of a longer word in B_s , when present, is a multiple of $h(s)$ and need not equal it. The interval must leave enough room for the costs of both the old divisors and the new words. More precisely, there are positive u_s, δ_s with

$$\begin{aligned} u_s - A_s &> \delta_s, \\ \rho(h(s)) - u_s - \delta_s &> \sup\{\epsilon_d : d \sqsubseteq h(s), d \in M \setminus P_s\}. \end{aligned} \quad (2.13)$$

Indeed, put $p = \rho(k(s)) - A_s \geq 0$ and $q = \rho(h(s)) - \rho(k(s)) > 0$. For $d \sqsubseteq k(s)$ missing P_s , every $x \in P_s$ misses d , and the old margin yields $\epsilon_d < \rho(k(s)) - \rho(x)$. The supremum of these costs is at most p . The supremum for $d \sqsubseteq h(s)$ not dividing $k(s)$ is strictly below q . Thus the full supremum on the right of Equation (2.13) is strictly less than $p + q = \rho(h(s)) - A_s$, also when $p = 0$. This leaves the required positive room for u_s and δ_s .

Define recursively on irreducibles

$$\begin{aligned} \rho(sY) &= u_s + \delta_s \rho(Y), & 0 < \epsilon_{sY} &\leq \delta_s \epsilon_Y, \\ \epsilon_{sY} &\leq \epsilon_{g(sY)} & \text{whenever the gate exists.} \end{aligned} \quad (2.14)$$

These choices give heights in $(0, 1)$ and costs in $(0, 1]$. We verify all comparisons, including the suprema.

For two elements in one block B_s , a divisor of the upper element from outside that block also divides the lower one. An old divisor belongs to P_s ; a new divisor with a different head must enter by its old gate, which then divides every member of B_s . Inside the block, cancellation removes s , and the margin follows by induction from the factor δ_s in both height differences and cost bounds.

Suppose an old x lies strictly below sY . Then $x \in P_s$. A missed divisor outside B_s is old in P_s , or has an old gate there of at least its cost. For such an old divisor d missing x , the join $x \vee d$ lies in P_s , and hence

$$\epsilon_d < \rho(x \vee d) - \rho(x) \leq A_s - \rho(x).$$

Their supremum is at most $A_s - \rho(x)$. Divisors inside B_s have cost at most δ_s . Equation (2.13) makes $\rho(sY) - \rho(x)$ strictly larger than both bounds.

Suppose sY lies strictly below an old y . Then $h(s) \sqsubseteq y$. Every divisor of y missing sY is old outside P_s , or has a gate there with at least its cost: a gate in P_s would put the divisor below all of B_s . Among these old costs, those for divisors of $h(s)$ are bounded by Equation (2.13). If $y \neq h(s)$, the remaining costs have supremum strictly below $\rho(y) - \rho(h(s))$ by the old margin. The total gap $\rho(y) - \rho(sY)$ is the sum of two positive gaps and exceeds both cost suprema. This proves this comparison case.

Distinct comparable new blocks have an intervening old gate, so the preceding cases apply transitively. To justify this use, if $x < z < y$, any divisor of y missing x either divides z and misses x , or misses z ; the sum of the two positive height gaps strictly dominates both relevant suprema.

Finally, for old $x < y$, any new divisor d of y missing x has an old gate $g(d) \sqsubseteq y$ missing x , with $\epsilon_d \leq \epsilon_{g(d)}$. No new cost increases the old supremum, while every old divisor remains present. The supremum is therefore exactly unchanged. This also proves persistence of the margins at unions of a conservative tower of these extensions. $\square$

2.4 The extension that merges finitely many copies

Lemma 2.5. *Let M satisfy the algebraic and margin hypotheses, and let a finite nonempty overlap prescription on copies of M be valid. There is a conservative extension realizing it by left translations. The extension is left cancellative and conical, has conditional joins, and admits extending heights and costs satisfying Equation (2.1). No new element divides an old one.*

Proof. Let D be the prescribed quotient of the copies, writing $t_i x$ for the image of x in copy i . This quotient carries one layer of the required overlaps; strings of its elements will complete that layer to a monoid. It is a right M -set and each map $x \mapsto t_i x$ is injective. On D use the order $z \sqsubseteq w$ when $zq = w$ for some $q \in M$; left cancellation and conicality inside a representing copy make this a partial order. Its right cones have conditional joins. Within one copy this is the old join. For elements represented in different copies, every common multiple must lie in their specified overlap, if there is one. In each copy join the element with the start of that overlap; transfer the two resulting elements into the overlap and join there. A missing join at any step means that the original cones are disjoint. This calculates exactly their intersection, because every common multiple must pass through both of those preliminary joins.

Adjoin generators l_i with the specified relations $l_i a_{ij} = l_j a_{ji}$. The normal forms are

$$xz_1 \cdots z_j \quad (x \in M, z_i \in D), \quad (2.15)$$

including $j = 0$. Each relation acts inside a new-generator/old-factor pair, exactly as in D ; it changes neither the number of new letters nor the preceding old coefficient. This proves uniqueness of the forms. Multiplication acts by the leading old coefficient of the second word on the last D -letter of the first, and then appends the remaining letters. For a fixed $z = t_i a$, the map $y \mapsto zy$ is

injective by left cancellation in copy i . Together with old left cancellation, this proves cancellation for arbitrary products of forms. Letter count proves conicality and that no new word divides an old element.

Unlike the probe extension, this construction never consumes a new letter. Its cones are therefore read from the final letter rather than from the first-letter blocks used above. For a positive-length word, right multiplication fixes the prefix before its final letter, advances that letter within its D -cone, and possibly appends further letters. More explicitly, compare $xz_1 \cdots z_p$ and $yt_1 \cdots t_q$ with $p, q > 0$. A common multiple requires $x = y$ and equality of the letters before position $\min(p, q)$. If $p = q$, their intersection is obtained by joining z_p and t_p in D , or is empty if that join is absent. If $p < q$, a common multiple exists exactly when z_p divides t_p in the right M -set D ; then the first word divides the second and the second is their join. The case $q < p$ is symmetric. An old x and a positive-length word with head y have a common multiple exactly when $x \sqsubseteq y$, in which case x divides that word. Two old words use their old join, because the heads of common multiples are precisely their old common multiples. These descriptions prove conditional joins and preserve old comparisons, quotients and nonjoins. The words $l_i x$ have exactly the identifications in D , as required.

It remains to supply margins. If the number of copies is r , set

$$\rho_D(z) = \frac{1}{r} \sum_{i \leq r} \rho_i(z), \quad \rho_i(z) = \begin{cases} \rho(x), & z = t_i x, \\ 0, & z \text{ has no representation in copy } i. \end{cases}$$

Each coordinate is nondecreasing along a comparison, and some coordinate increases strictly in a proper comparison. Choose positive costs with

$$\epsilon_z^D \leq \frac{1}{r} \min(\{\rho(1)/2\} \cup \{\epsilon_x : z = t_i x \text{ for some } i\}).$$

For $u < v$ in D , classify a divisor d of v missing u by a copy representing d . That copy also represents v . If it represents u , its old margin bounds the supremum of these costs strictly below its coordinate increase divided by r . Otherwise that increase is at least $\rho(1)$, while each such cost is at most $\rho(1)/(2r)$. There are only finitely many copies, so their strict bounds still give the required strict supremum bound for $\rho_D(v) - \rho_D(u)$.

Choose $\theta, \phi > 0$ with $\theta + \phi < 1$, and recursively set

$$\begin{aligned} \rho(wz) &= \rho(w) + (1 - \rho(w))(\theta + \phi \rho_D(z)), \\ 0 < \epsilon_{wz} &\leq (1 - \rho(w)) \min(\theta/2, \phi \epsilon_z^D). \end{aligned} \tag{2.16}$$

For comparable new words of equal length only their last letters vary; missed divisors of that length are handled by the scaled D -margin, and all shorter divisors divide both words. For a word w compared with wz , every missed divisor has the new length and prefix w , so its cost is at most $(1 - \rho(w))\theta/2$, strictly less than the height increase. An arbitrary comparison factors through these comparisons, and, when the lower word is old, an old comparison of leading coefficients. Transitivity of the margin, proved in Lemma 2.4, finishes the check. Old cost suprema are unchanged because there are no new old divisors. $\square$

2.5 Conservation under simulation

We next establish the compatibility needed to reorder the construction. This step also explains why overlap prescriptions remain valid when operations are scheduled before they are executed.

Lemma 2.6. *Suppose $B \subseteq M$ is conservative and both monoids have the algebraic properties above. Either extension operation with parameters in B is valid over M , and the extension of B conservatively embeds into the corresponding extension of M by the same generators. A finite overlap prescription with parameters in B is valid in B if and only if it is valid in M .*

Proof. For the last assertion, the only nontrivial transitivity check passes through three copies. In the middle copy, intersect the two prescribed overlap starts. If their old cones are disjoint, conservativity keeps them disjoint. Otherwise their old join and its right quotients are unchanged. Every later common multiple is a multiple of that join. The two endpoint identifications at the join already agree in a valid prescription over B , so right multiplication gives agreement at all later common multiples. This proves validity over M . Conversely, restrict a valid prescription to B . Comparisons between its old parameters have their quotients in B , so no identification or transitivity witness between elements of B requires leaving B . The restriction has exactly the indicated pair relations and is valid.

For probes, equality of two letters in the smaller direct limit is preserved and reflected because it is an equality at a common finite stage. If an old $b \in B$ divides a smaller letter $[x, i]$ in the larger direct limit, there is a common stage at which

$$by = xd_i \cdots d_{j-1}.$$

Both endpoints of this comparison belong to B , so its right factor y belongs to B . Thus divisibility of a smaller letter by an old smaller multiplier, including its quotient, is unchanged. The same irreducible words therefore embed. The gate recursion uses only the previous gates and old joins and quotients, so it has the same outcome, including the absence of a gate. The join calculations in Lemma 2.3 now preserve and reflect joins and nonjoins.

For clarity, preservation of right quotients between new forms follows by a separate division argument. If both forms are old, use the base hypothesis. If the divisor is old and the dividend new, divide its first direct-limit letter by the preceding common-stage calculation, and retain its suffix. If both forms have the same new head, cancel that head and recurse on their suffixes. In the remaining cases the new divisor must leave its first-letter block. Factor through its gate, using the quotient in Equation (2.11), and divide that old gate into the dividend by the first two cases. Every factor obtained lies in the smaller extension. Left cancellation makes their product the unique required quotient.

For mergers, representation in an overlap and intersections of cones in D were calculated solely using old joins and their right quotients. These calculations are consequently unchanged. The normal forms in Equation (2.15) then preserve comparisons, their right factors, joins and nonjoins. This proves conservative simulation for both operations. $\square$

Lemma 2.7. *Suppose a probe satisfies Equation (2.6) in one stage of a tower of the two operations. It continues to satisfy that equation at every later stage and at the union, with the full later downward closure of its original path. Previously constructed joins between probes are also preserved.*

Proof. A merger introduces no new divisor of the old a . At a probe extension, a new $x \sqsubseteq a$ has an old gate $g(x) \sqsubseteq a$. If x also divides the old probe, its gate lies below that probe and hence below some original path element; then x does too. If x does not divide the old probe, its gate does not either. The old probe cannot divide x , because it does not divide a . The join calculation through the gate therefore gives ae in this case. This proves both alternatives at a successor step. At a limit every tested element has already appeared, and all comparisons and joins have been preserved. Conservativity directly preserves joins between existing probes. $\square$

2.6 The schedule and the countable-base rearrangement

We now assemble the operations. The height bound ensures that the apparently arbitrary ideal requirements can be scheduled using only countable paths. The rearrangement proved afterward concerns the same resulting monoid, not a separately constructed substitute.

Proof of Theorem 2.2. Saturating the geometric requirements. Start with the free monoid on e . For $n < \omega$, for example, put

$$\rho(e^n) = 1 - 2^{-n-1}, \quad \epsilon_{e^n} = 2^{-n-3}.$$

These satisfy Equation (2.1). Perform ω_1 rounds. At the start of a round schedule every available probe datum $(a, b, (n_i))$ with the fixed e , and every available finite valid merger. For each probe datum schedule μ successive insertions. Execute the resulting wellordered list with the two lemmas, and take unions at limits. Lemma 2.6 keeps every scheduled merger valid until it is executed.

A monoid of size at most μ has at most $\mu^{\aleph_0} = \mu$ countable paths and at most μ finite prescriptions. A probe extension uses a quotient of $M \times \omega$ and finite strings; a merger uses finitely many copies and finite strings. Thus every operation preserves size at most μ , and each round has at most μ steps. The union of ω_1 such rounds still has size at most μ . The margin lemmas ensure that old values are never changed and that old missed-cost suprema never increase, so the final union T has Equation (2.1) as well as all the algebraic properties.

Given an ideal in $\downarrow_T a$, use Lemma 2.1 to obtain a cofinal path starting at 1. Its elements, a, b , and all its transition and terminal quotients occur before one round: the set is countable and the rounds have cofinality ω_1 . Alternatively, quotient conservativity already puts every quotient into any stage containing its endpoints. The corresponding requirement is scheduled. Its final lower closure is the given ideal, and Lemma 2.7 therefore gives Equation (2.2) for the μ insertions. For any earlier and later insertion with the same datum, the earlier probe is old but is not a divisor of a , so it is not an old divisor of the new probe. Joining through the new probe's gate gives ae . This join persists in the final union, proving independence.

Every final finite overlap prescription has its parameters in one stage. Its validity reflects to that stage by Lemma 2.6, and the schedule realizes it later. The realization remains exact by conservation of comparisons and quotients: equality $l_i x = l_j y$ in a later stage must still factor through the preserved join of the two roots, with its prescribed quotients. Thus later right factors add exactly the stipulated identifications and no others. Finally use $a = 1, b = e$ and $U = \{1\}$. Its probes have no old divisor e , so there are μ elements of $T \setminus eT$. This proves both cardinality assertions.

Extracting a countable initial base. It remains to prove the local construction property for this same T . We first select countably many operations containing any prescribed countable set of elements, and then perform all omitted operations over their union. The resulting tower must retain the gate rule: this is what will let a later descent replace a position by one born at an earlier stage. Regard the entire schedule as a continuous tower $(M_\alpha)_{\alpha \leq \lambda}$ of individual operations. Record each parameter by a finite word in previous generators. Every element of the presented monoid has such a word, and each equality has a finite derivation from the relations. This does not assert that a whole countable path has finite support.

Choose a sufficiently large regular Θ and a countable elementary submodel N of H_Θ containing the tower, its presentations and an enumeration of the prescribed countable subset of T . Countable objects in N , in particular the parameter paths of operations indexed in N , have all their entries in N . First perform just the operations whose indices belong to N , in their original order. Let S_α denote this subconstruction using selected indices below the cut α .

At cuts $\alpha \in N$ we have

$$S_\alpha = M_\alpha \cap N, \quad (2.17)$$

with a conservative inclusion in M_α . Here is the induction justifying the assertion. Old comparisons, their quotients and joins in an elementary intersection are reflected by elementarity; a nonjoin remains a nonjoin in the larger monoid. Before a selected operation, every parameter is already present: its finite term code belongs to N , and all entries of a countable path in N belong to N . Conservative simulation makes that operation legal in the subconstruction and embeds its result. Conversely, every element of $M_\alpha \cap N$ has by elementarity a finite term code in N ; all of its finitely many generator indices belong to N and precede α . It is therefore produced by the selected subconstruction. This proves Equation (2.17) and the successor step of the induction. At a cut $\alpha \notin N$, fix the finitely many words in a comparison or join test. If they use selected operation indices, let γ be the greatest such index and put $\beta = \gamma + 1$; if there are no such indices, put $\beta = 0$. Then $\beta \in N$, $\beta < \alpha$, and all the words belong to $S_\beta = M_\beta \cap N$. The original inclusion $M_\beta \subseteq M_\alpha$ is conservative. Any right quotient or conditional join required by the test therefore already lies in M_β and, by elementarity, in N , hence in S_β . This also preserves nonjoins and proves conservation at α . Unions introduce no additional finite relation. This completes the induction.

The resulting selected submonoid T_0 is countable: the initial monoid is countable, there are countably many selected steps, and each operation over a countable base adds only countably many elements. It contains the prescribed sequence by its finite term codes. The set of selected indices need not be an initial segment; the argument has used Equation (2.17) only at cuts belonging to N .

Reinserting the omitted operations. Append now the omitted steps in their original order. Before an omitted step α , the current presentation is the union of the original relations below α and all selected relations. It has an alternative realization: start with M_α , and then perform the selected steps at indices at least α in their order. Indeed $S_\alpha \subseteq M_\alpha$ is conservative, and repeatedly simulating the selected subconstruction makes all these operations legal. Each extends its whole base conservatively, so this alternative realization conservatively contains M_α . It has precisely the same generators and relations as the current reordered presentation. Consequently the original operation at α is legal over the current base, by conservative simulation again. Induction validates every appended step. At limits the presentations are their unions; at the end their union is exactly the original presentation of T .

Each step of this reordered tower is one of the two operations. The merger has no new old divisors, and the probe extension has its least-old-multiple rule, including Equation (2.3). Conservation allows those comparisons to be tested in T . This proves the last assertion and completes the theorem. $\square$

We isolate the consequence of the local construction property that will be used for symmetries. It makes explicit why a position below a countable path can be moved to the countable base after finitely many steps; it does not claim that every such position was in that base already.

Lemma 2.8. *Let (n_i) be a nondecreasing sequence in T , and choose a countable-base tower from Theorem 2.2 whose initial submonoid contains all n_i and all right quotients between them. Let $\mathcal{A} \subseteq \omega \times T$ be a set of pairs such that each $(l, m) \in \mathcal{A}$ has $m \sqsubseteq n_N$ for some N . Assume that for each $(l, m) \in \mathcal{A}$ and each N there is $(l', md) \in \mathcal{A}$ with $l' \geq l$, $n_l d = n_{l'}$, and $n_N \sqsubseteq md$. Then, starting from any pair in $\mathcal{A}$, a finite succession of these replacements reaches a pair whose second coordinate belongs to the initial countable submonoid.*

Proof. If m is not in that submonoid, take its first birth step in the reordered tower. It has the old multiple n_N , so this is a probe step and its gate satisfies $g(m) \sqsubseteq n_N$. Choose an admissible replacement with $n_N \sqsubseteq md$. The quotient d belongs to the initial submonoid and is therefore old at this step. Conservation and Equation (2.3) imply that md belongs to the strictly earlier base. Repeat if necessary. Each repetition strictly lowers the birth ordinal, so the process terminates after finitely many repetitions. The resulting position lies in the countable initial submonoid. $\square$

3 Product forcing and support locality

The monoid supplies copies of diagrams with controlled intersections. We now turn those copies into two locality principles. After the Cohen generic is fixed, a diagram decides statements whose name parameters it supports. A value represented on two independent diagrams can then be named on their common part. These principles will classify quotients and give the choice and coding properties needed to pass from quotients to arbitrary sets.

3.1 The choice ground and the two forcings

The product separates the generic labels from their ground-model diagram geometry. A related Cohen/tower product appears in [9, Section 3]; the monoid and the diagram order used here are specified below.

Work in a ground model V of ZFC. Set

$$\kappa = \aleph_1^V, \quad \mathfrak{c} = (2^{\aleph_0})^V, \quad \mu = (\mathfrak{c}^+)^V, \quad |I|^V = \mu^+.$$

In the constructible ground used for the consistency result, $\mathfrak{c} = \aleph_1$, so these parameters are exactly $\kappa = \aleph_1$ and $\mu = \kappa^+$. The slightly more general choice of μ will allow us to retain an arbitrary given transitive choice ground. All objects and cardinal calculations in the construction, unless explicitly placed in an extension, belong to V .

For every infinite cardinal $\lambda < \mu$,

$$\lambda^{\aleph_0} \leq \mathfrak{c}^{\aleph_0} = \mathfrak{c} < \mu.$$

Since μ is regular and uncountable, every countable sequence in μ is bounded there. It follows that

$$\mu^{\aleph_0} = \mu. \tag{3.1}$$

Take the monoid T , with distinguished $e \neq 1$, from Theorem 2.2, using this μ . Let

$$\mathbb{R} = \text{Add}(\kappa, I), \quad \mathbb{P} = \{r : T \hookrightarrow I\}, \quad \mathbb{Q} = \mathbb{R} \times \mathbb{P}.$$

Here a condition of $\mathbb{R}$ is a partial function $I \times \kappa \rightarrow 2$ with countable domain, ordered by reverse inclusion. Its coordinate support is

$$\text{supp}_I(d) = \{j \in I : (\exists \alpha < \kappa) (j, \alpha) \in \text{dom}(d)\}.$$

For $r \in \mathbb{P}$ and $n \in T$, put

$$r_n(x) = r(nx), \quad D_r = r[T], \quad r \leq_{\mathbb{P}} s \iff (\exists n \in T) s = r_n.$$

Thus a stronger diagram contains a specified copy of a weaker one. The point $r(1)$ is its center. Whenever $s = r_n$, the element n is determined by the centers, since $s(1) = r(n)$. Left cancellation

makes every r_n injective. If $r = r_{nm}$, injectivity and conicality give $n = m = 1$; this verifies antisymmetry of the order. Transitivity follows from $(r_n)_m = r_{nm}$.

The poset $\mathbb{P}$ always denotes this *ground* set, including when it is forced over the Cohen extension. No closure or cardinal-preservation assertion about $\mathbb{P}$ or the full product is needed.

Lemma 3.1. *The forcing $\mathbb{R}$ is countably closed and satisfies the μ -chain condition. It adds no countable sequences of ground elements, preserves $\kappa = \aleph_1^V$, and preserves every ground cardinal at least μ . In particular it preserves μ and μ^+ . If $V \models \text{CH}$, it preserves all cardinals.*

Proof. A descending countable sequence has its union as a lower bound. Deciding the entries of a name for a sequence in a ground set and then taking such a lower bound shows that no new countable sequence in that set is added. This also preserves ω_1^V .

We recall the relevant delta-system argument to identify the exact cardinal bound. A family of μ countable sets has a delta-system subfamily of size μ . Code its union into μ , and write its members as $S_\alpha \subseteq \mu$, for $\alpha < \mu$. There is a club of $\delta < \mu$ such that $S_\alpha \subseteq \delta$ whenever $\alpha < \delta$. On the stationary part of this club of cofinality ω_1 , each $S_\alpha \cap \alpha$ is bounded in α . The pressing-down lemma makes a bound constant, say $\gamma < \mu$, on a stationary subset. There are at most $|\gamma|^{\aleph_0} + \aleph_0 < \mu$ possible countable subsets of γ , by Equation (3.1) and its preceding bound. Regularity of μ therefore makes $S_\alpha \cap \alpha$ one fixed set on a stationary subset. For $\alpha < \beta$ in that subset, $S_\alpha \subseteq \beta$, so their intersection is this fixed root.

Apply this to the domains of μ Cohen conditions. There are at most $2^{\aleph_0} = \mathfrak{c} < \mu$ assignments on the countable root. Thinning once more gives two, indeed μ , conditions agreeing on the root. They are compatible, which proves the chain condition. The usual antichain bound then preserves all cardinals at least μ : the possible values at any one coordinate of a name for an ordinal-valued function are covered by fewer than μ ground ordinals, and regularity handles the case of μ itself. Countable closure and the chain condition together preserve all cardinals when $\mu = \aleph_2$. In a general ground we assert no preservation of cardinals strictly between κ and μ .

Finally, any sequence of ground elements has its values in some ground rank segment: their ranks have an ordinal bound, and forcing preserves ordinals. Thus the assertion about sequences in ground sets covers all countable sequences of ground elements. $\square$

Lemma 3.2 (Extending and amalgamating diagrams). *1. For every $p \in \mathbb{P}$ and ground $J \subseteq I$ of size at most μ , there is $r \in \mathbb{P}$ with $r_e = p$ and $J \subseteq D_r$.*

2. A finite family $s_i \in \mathbb{P}$ has a common extension whenever each pair has either disjoint ranges or an intersection that is exactly the range of a common subdiagram

$$s_{i,a_{ij}} = s_{j,a_{ji}}.$$

3. A partial placement of finitely many diagrams that is injective and consistent on their specified common parts, extends to a ground permutation of I . Unspecified coordinates of these diagrams and of finitely many Cohen conditions can be placed outside any prescribed ground set of size at most μ .

Proof. The probes from Theorem 2.2 for $a = 1, b = e, U = \{1\}$ give μ distinct elements m with $me = e$. None belongs to eT : if $m = ex$, left cancellation would give $xe = 1$, contrary to conicality and $e \neq 1$. Hence $|T \setminus eT| = \mu$. Define $r(ex) = p(x)$, using left cancellation, and fill the complement injectively in $I \setminus D_p$, including $J \setminus D_p$. Both the size of that complement in T and the strict inequality $\mu < |I|$ give the first assertion.

For the second assertion, identify the copies of T exactly where their labels in I coincide. The hypothesis says that these identifications have the form required by the finite-merger property; their transitivity and absence of collapse within a copy follow from actual equality of labels. Let $l_i \in T$ realize this merger. On $\bigcup_i l_i T$, assign $l_i x$ the value $s_i(x)$. The exact-overlap property makes this well defined and injective. Extend it injectively to the rest of T . The resulting r satisfies $r_{l_i} = s_i$.

For the last assertion, the prescribed domain and range each have size at most μ . Their complements in I both have cardinality μ^+ , so a bijection between the complements extends the partial bijection to a permutation. Fresh images can first be chosen off the excluded set, whose size is still at most μ , and then the same argument applies. All these choices occur in V . $\square$

3.2 The symmetric system and transport

Let $G = \text{Sym}(I)^V$. A permutation acts simultaneously on diagrams and on Cohen coordinates:

$$(\pi r)(x) = \pi(r(x)), \quad (\pi d)(\pi j, \alpha) = d(j, \alpha).$$

These are automorphisms of $\mathbb{Q}$. For ground $J \subseteq I$, let $\text{Fix}(J)$ be its pointwise stabilizer in G , and set

$$\mathcal{F} = \{K \leq G : (\exists J \in V) [J \subseteq I, |J|^V \leq \mu, \text{Fix}(J) \subseteq K]\}.$$

This is a normal filter of subgroups. Finite intersections are handled by $J \cup J'$, and $\pi \text{Fix}(J) \pi^{-1} = \text{Fix}(\pi J)$.

The action on names is recursive:

$$\pi \tau = \{(\pi \sigma, \pi u) : (\sigma, u) \in \tau\}.$$

A name has support J when $\text{Fix}(J)$ fixes it literally. A hereditarily symmetric name, abbreviated HS, has a support and only hereditarily symmetric member names. There is no restriction on name rank. Ground check names are fixed by every permutation. Canonical ordered-pair and sequence names commute with the action. When the poset has no largest condition, their usual definitions use all conditions in place of that condition; equivalently one may adjoin a fixed largest condition.

For the evaluated presentation, assume that V is externally transitive and that the indicated generic filters are available in the ambient universe. Over a general choice ground, the same construction and the generic notation below express internal forcing assertions; they do not require external recursion on its membership relation. Section 8 realizes these assertions over an arbitrary countable ground by a quotient of internal names.

Write F for an $\mathbb{R}$ -generic filter over V , and H for a generic on the ground $\mathbb{P}$ over $V[F]$. The product extension is $V[F][H]$. A product name τ , after evaluating its Cohen part, becomes a $\mathbb{P}$ -name denoted by $\tau[F]$. Let

$$W = \{\tau^{F \times H} : \tau \in \text{HS}^V\}.$$

Theorem 3.3 (The symmetric-model theorem). *The symmetric forcing relation is definable formula by formula in V , and it forces every axiom instance of full pure-set ZF. If V is externally transitive, the evaluated class W is a transitive model of that theory, with*

$$V \subseteq W \subseteq V[F][H], \quad \text{Ord}^W = \text{Ord}^{V[F][H]}.$$

The truth lemma holds for this evaluated model. Over an arbitrary countable choice ground, it holds for the symmetric quotient of internal names described in Section 8.

Proof. Apply the set-forcing symmetric-extension theorem [11, Section 2.1, p. 4] to $\mathbb{Q}$, G , and the normal subgroup filter verified above. Its internal formulation concerns all hereditarily symmetric names and yields the full Separation and Replacement schemata, as well as Foundation; this formulation is recalled in Lemma 8.2. It requires no countable-completeness assumption on the filter. External transitivity of V permits the usual recursive name evaluation and gives the displayed literal inclusions. For an arbitrary countable ground, Lemma 8.3 instead gives truth in the membership-closed symmetric quotient; it makes no assertion of external well-foundedness. $\square$

The generic terminology below expresses arguments with the ordinary and symmetric forcing relations. In particular, each assertion about all generics can instead be read as its forcing-theoretic assertion over the choice ground. The formal consistency deduction is given after the construction.

We will use symmetry in its precise product form:

$$(d, s) \Vdash_{\mathbb{Q}} \varphi(\vec{\tau}) \iff (\pi d, \pi s) \Vdash_{\mathbb{Q}} \varphi(\pi \vec{\tau}). \quad (3.2)$$

The corresponding evaluation identity is

$$(\pi \tau)^{\pi F \times \pi H} = \tau^{F \times H}.$$

It does not define an automorphism of one fixed extension by sending $\tau^{F \times H}$ to $(\pi \tau)^{F \times H}$. The product/iteration equivalence also gives the factorized version of Equation (3.2): a Cohen condition forcing $s \Vdash_{\mathbb{P}} \varphi(\vec{\tau}[F])$ is transported to a Cohen condition forcing the corresponding assertion at πs with names $\pi \vec{\tau}$. Here πs is explicitly reindexed ground data; it is not the image of $\check{s}$ under the name action, which fixes check names.

Let $\dot{A}_j$ be the canonical name for the Cohen subset of κ at coordinate j , and put

$$A_j = \dot{A}_j^{F \times H}, \quad A = \{A_j : j \in I\}.$$

The name $\dot{A}_j$ has support $\{j\}$, and $\pi \dot{A}_j = \dot{A}_{\pi j}$. The name for A is invariant. Distinct coordinates give distinct labels: below any Cohen condition, assign different bits at one ordinal unused at the two coordinates. This proves forced distinctness, which persists after the second forcing.

3.3 Decision and descent

Restriction to supporting coordinates and descent to a common support have precedents in [9, Lemmas 10 and 15]. The following proofs establish the versions needed for our diagrams, including arbitrary-rank product names and exact compatibility of the moved conditions.

The two factors play different roles in the first lemma. Once the Cohen generic is fixed, strengthening a supporting diagram cannot change the truth of a supported assertion. The assertion can still depend on the Cohen generic; the second part says that a Cohen condition witnessing it may discard coordinates outside the supporting diagram.

Lemma 3.4 (Decision over a supporting diagram). *Fix an $\mathbb{R}$ -generic F .*

1. *If $p \in \mathbb{P}$ and all product-name parameters $\tau_1, \dots, \tau_k$ are supported by D_p , then p decides each ordinary $\mathbb{P}$ -forcing formula $\varphi(\tau_1[F], \dots, \tau_k[F])$ in $V[F]$. The parameters may have arbitrary rank.*
2. *Suppose a Cohen condition d forces that $s \Vdash_{\mathbb{P}} \varphi(\vec{\tau}[F])$, where all the product names are supported by D_s . Then $d \restriction (D_s \times \kappa)$ forces the same assertion.*

For brevity we write $d \upharpoonright D_s$ for this restriction.

Proof. If the first assertion fails, there are $s_0, s_1 \leq p$ forcing contrary answers in $V[F]$. Choose $d_0, d_1 \in F$ witnessing those answers in the factorized forcing relation. Their restrictions to $D_p \times \kappa$ are compatible. By Lemma 3.2, choose $\pi \in \text{Fix}(D_p)$ so that, outside D_p , the coordinates of s_1, d_1 move away from those of s_0, d_0 . The diagrams $s_0, \pi s_1$ meet exactly in their common subdiagram p , so have a common extension. The conditions $d_0, \pi d_1$ agree on the root and have disjoint supports off it, so are compatible. All parameter names are fixed by π . Equation (3.2) now gives two contrary assertions forced by compatible product conditions, a contradiction. This argument does not assert that πd_1 belongs to F .

For the second assertion, put $\delta = d \upharpoonright D_s$. If δ does not force the indicated factorized assertion, there is $c \leq \delta$ forcing its negation. Choose $\pi \in \text{Fix}(D_s)$ moving $\text{supp}_I(d) \setminus D_s$ off $\text{supp}_I(c) \setminus D_s$. The condition πd is compatible with c . It forces the original factorized assertion, by product symmetry, because s and all its parameter names are fixed. This contradicts the assertion forced by c . $\square$

We next show that equality between independently supported values can be represented on the intersection. The proof uses a third copy to compare the two values while keeping an arbitrary potential counterexample compatible.

Lemma 3.5 (Support descent). *Suppose $s_0, s_1 \leq p$ and $D_{s_0} \cap D_{s_1} = D_p$. Let σ_i be product HS names supported by D_{s_i} . Fix an $\mathbb{R}$ -generic F . If some $v \leq s_0, s_1$ forces $\sigma_0[F] = \sigma_1[F]$ in ordinary $\mathbb{P}$ -forcing, then there is a product HS name σ , supported by D_p , such that*

$$s_i \Vdash_{\mathbb{P}} \sigma[F] = \sigma_i[F] \quad (i = 0, 1)$$

in $V[F]$. The names may have arbitrary rank. If, in $V[F]$, each s_i also forces $\sigma_i[F] \subseteq A$, the descended name can be normalized to a subset name with only label names as possible members.

Proof. Take $d \in F$ witnessing the equality forced at v . Both parameter supports lie in D_v , so Lemma 3.4 allows us to replace d by $d \upharpoonright D_v$. Set $\delta = d \upharpoonright D_{s_0}$. We will combine the translates of σ_0 into one D_p -supported name. A translate contributes its value only when the generic contains the corresponding translate of (δ, s_0) ; the value of the new name will be the union of these contributing values. Thus the main task is to prove that any two contributing copies agree. For this coherence argument, allow both factors of the product generic to vary. We will return to the fixed F after constructing the name.

For every $\pi \in \text{Fix}(D_p)$, call the copy $\pi\sigma_0$ *active* when its trigger $(\pi\delta, \pi s_0)$ belongs to the product generic. We first prove that any two active copies have equal values.

Consider two copies whose diagrams $\bar{s}, \bar{s}'$ have intersection exactly D_p . If they can be unequal while active, some product condition (c, t) extends both triggers and forces the inequality. Place a copy u of s_1 over p , with the rest of its range fresh relative to $D_t \cup \text{supp}_I(c)$. Extend the respective placements of (s_0, s_1) onto $(\bar{s}, u)$ and $(\bar{s}', u)$ to placements of v , producing diagrams v', v'' . Use the same placement on all of D_{s_1} in both cases. Choose all remaining coordinates freshly, so that the ranges have the intersections shown in Figure 2:

$$D_{v'} \cap D_t = D_{\bar{s}}, \quad D_{v''} \cap D_t = D_{\bar{s}'}, \quad D_{v'} \cap D_{v''} = D_u. \quad (3.3)$$

Every intersection in this display is its indicated common subdiagram. The required partial maps agree on the root, and their domain and range sizes are at most μ ; Lemma 3.2 therefore supplies the ground permutations realizing these placements.

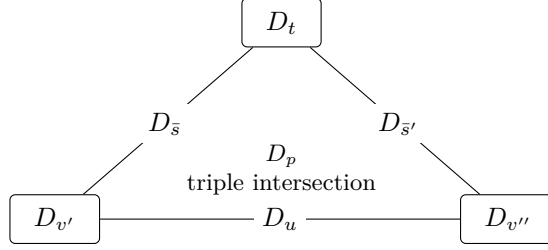


Figure 2: The independent-copy case of support descent. Each edge is labelled by the exact intersection of its endpoint ranges. The two placed equality witnesses v', v'' use the same fresh copy u of s_1 ; t carries the proposed inequality between the copies on $\bar{s}$ and $\bar{s}'$, whose ranges meet exactly in D_p . These exact intersections allow the three diagrams to be amalgamated over their common root p .

The two placed copies of d agree on D_u . They agree with c on $D_{\bar{s}}$ and $D_{\bar{s}'}$, respectively, because c extends the two placed copies of δ . On the remaining coordinates freshness gives compatibility. Thus the Cohen conditions have a common extension. The three diagrams v', v'', t also have a common extension by Equation (3.3) and finite amalgamation. The two transported σ_1 names are literally equal, since the placements agree on the entire D_{s_1} . The equalities transported from (d, v) therefore contradict the inequality forced by (c, t) .

For two active copies with arbitrary overlap, start again with any proposed inequality-forcing common extension (c, t) . Place a third copy of s_0 over p , independent of t off the root, and place its Cohen trigger compatibly with c . It has a common product extension with (c, t) . The independent case just proved applies between this third copy and each of the original ones, contradicting the inequality. This proves coherence of all active copies, below every possible inequality witness.

Define a mixture using all joint strengthenings:

$$\sigma = \left\{ (\pi\eta, (c, t)) : \begin{array}{l} \pi \in \text{Fix}(D_p), \quad (\eta, (a, u)) \in \sigma_0, \\ (c, t) \leq (\pi\delta, \pi s_0), \quad (c, t) \leq (\pi a, \pi u) \end{array} \right\}. \quad (3.4)$$

This is a set name. Left multiplication in $\text{Fix}(D_p)$ permutes its entries, so it is supported by D_p . Its member names are translates of member names of σ_0 and hence are HS. In any product generic its value is the union of the values of the active copies. Both inclusions in this description follow from directedness of the generic: all common strengthenings, rather than a selected one, occur in Equation (3.4). When the original trigger is active, coherence makes this union exactly the value of σ_0 . Return now to the fixed F . Since $\delta \in F$, the original trigger is active in every product generic with this Cohen factor and with s_0 in its diagram filter. It follows that $s_0 \Vdash_{\mathbb{P}} \sigma[F] = \sigma_0[F]$. The equality with $\sigma_1[F]$ holds below v . Its parameters are supported by D_{s_1} , so Lemma 3.4 makes it hold already below s_1 .

No step imposed a rank bound. For the subset assertion, replace the mixture by its intersection with A , using the normalization in the next paragraph. $\square$

For $r \in \mathbb{P}$, define the following ground set of names:

$$\mathcal{X}(r) = \left\{ \tau \subseteq \{\dot{A}_j : j \in I\} \times \mathbb{Q} : (\forall \pi \in \text{Fix}(D_r)) \pi\tau = \tau \right\}. \quad (3.5)$$

Every member is HS. If a product HS name τ is supported by J , define

$$\tau \restriction A = \{(\dot{A}_j, u) : j \in I, u \in \mathbb{Q}, u \Vdash_{\mathbb{Q}} \dot{A}_j \in \tau\}.$$

The ordinary truth lemma gives value $\tau^{F \times H} \cap A$. Symmetry preserves the support J . In particular a name supported by D_r whose actual value is a subset of A has the same value as some member of $\mathcal{X}(r)$. This is a set-sized normalization, valid for an original name of any rank.

3.4 Coding arbitrary sets by the labels

The next argument extracts a partial function from an orbit of a name. The diagram condition in the function's definition ensures that equal inputs have equal values.

Proposition 3.6. *In W , every nonempty set X is the image of a surjection $A \times \eta \rightarrow X$ for some ordinal η . No bound on the rank of X is imposed.*

Proof. Let σ be a ground HS name, and let p have D_p as a support for σ . Form the graph name

$$\dot{h}_{\sigma,p} = \left\{ \left(\text{op}(\dot{A}_{\pi(p(1))}, \pi\sigma), (1_{\mathbb{R}}, \pi p) \right) : \pi \in G \right\}, \quad (3.6)$$

where op denotes the canonical ordered-pair name. This name is invariant, and all its member names are HS.

Its value is a partial function on A . Indeed, suppose two contributing pairs have equal first coordinates. Label distinctness gives $\pi(p(1)) = \rho(p(1))$. There is a common extension $r \in H$ of $\pi p, \rho p$; write $\pi p = r_n$ and $\rho p = r_m$. The equality of centers and injectivity of r give $n = m$. Consequently $\pi p = \rho p$ pointwise, so $\rho^{-1}\pi \in \text{Fix}(D_p)$. It fixes σ , and hence $\pi\sigma = \rho\sigma$ literally. The two values therefore agree. This argument concerns names under compatible diagram conditions, not an action on one fixed symmetric model.

The orbit construction gives invariant partial functions. We now form a set-indexed family of them whose ranges cover X . Choose an HS name τ for X . The set M of member names appearing in τ is a ground set of HS names. In V , enumerate by an ordinal η the set

$$B = \{(\sigma, p) \in M \times \mathbb{P} : \text{Fix}(D_p) \text{ fixes } \sigma\}.$$

Every member name has a support of size at most μ , and Lemma 3.2 provides supporting diagrams. We include all such diagrams, because one chosen diagram need not belong to the generic filter. The corresponding sequence of graph names $\dot{h}_{\sigma,p}$ is a ground sequence of invariant HS names. Its canonical sequence name is itself invariant and HS, so its interpretation $\langle h_i : i < \eta \rangle$ belongs to W .

These partial functions cover X . For $x \in X$, an active member name σ of τ has value x . The set of diagrams whose ranges contain a fixed support of σ is ground dense, by Lemma 3.2. Thus some such p belongs to H . The identity entry of $\dot{h}_{\sigma,p}$ maps $A_{p(1)}$ to x .

All remaining definitions take place in W . For nonempty X , fix one $x_0 \in X$, and define

$$h(a, i) = \begin{cases} h_i(a), & a \in \text{dom}(h_i) \text{ and } h_i(a) \in X, \\ x_0, & \text{otherwise.} \end{cases}$$

Separation and Replacement form this function on $A \times \eta$, and coverage makes it surjective. The one default element uses only nonemptiness of X . The enumeration above is over $M \times \mathbb{P}$, a set depending on the particular name τ ; no restriction to names below a fixed rank was made. $\square$

3.5 Choice for ordinal-indexed families

The presentation just proved reduces arbitrary families to families of subsets of A . We first find one wellorderable part of A meeting every member of a given ordinal-indexed family of nonempty subsets.

Proposition 3.7. *The model W satisfies AC_{WO} : every ordinal-indexed family of nonempty sets in W has a choice function in W .*

Proof. We first treat a family $\langle X_\xi : \xi < \eta \rangle$ of nonempty subsets of A . Start below a product condition (d_0, p) forcing this assertion for an HS family name. By extending the diagram, arrange that D_p contains its support and $\text{supp}_I(d_0)$. The assertions that a given graph is a function and has nonempty subset values are absolute between W and the full extension. Consequently this use of ordinary forcing follows equally from a symmetric-forcing hypothesis. Names for the individual X_ξ can be obtained by the membership forcing relation from the family and $\check{\xi}$; they retain support D_p . Choose once and for all q with $q_e = p$. This same q will work for every family index. Witnesses already available on p need no move; for witnesses on proper extensions of p , the probes place independent copies inside q and Cohen genericity activates one of them.

Fix F containing d_0 , and fix $\xi < \eta$. In $V[F]$, some $t \leq p$ and $j \in I$ satisfy

$$t \Vdash_{\mathbb{P}} A_j \in X_\xi.$$

This follows by deciding a witness in the ground index set I for membership in A . Extend t to include j in its range, and write $p = t_b$. Choose $d \in F$ witnessing the displayed forcing assertion, and restrict it to D_t by Lemma 3.4. If $b = 1$, then $t = p$, so q already forces a witness whose index belongs to D_q .

Suppose $b \neq 1$. The probes of Theorem 2.2, with $a = 1, U = \{1\}$, give μ elements m satisfying $mb = e$ and $mT \cap m'T = eT$ for distinct probes. Thus

$$(q_m)_b = p, \quad D_{q_m} \cap D_{q_{m'}} = D_p \quad (m \neq m').$$

Each q_m is a copy of t over p . Choose in V a permutation $\pi_m \in \text{Fix}(D_p)$ taking t to q_m . The conditions $\pi_m d$ all restrict on D_p to $d_* = d \restriction D_p$, and their coordinate supports off that root are pairwise disjoint.

For these fixed ground data, the set

$$\{c : c \perp d_*\} \cup \{c : (\exists m) c \leq \pi_m d\} \tag{3.7}$$

is dense in $\mathbb{R}$. Given c compatible with d_* , first extend it by d_* . Its support is countable, so it meets the nonroot support of only countably many of the pairwise disjoint copies. Choose a remaining probe and extend c by its placed condition $\pi_m d$. This proves density. Because $d_* \in F$, the generic meets the second alternative of Equation (3.7). For the resulting m , product symmetry and $\pi_m X_\xi = X_\xi$ give

$$q_m \Vdash_{\mathbb{P}} A_{\pi_m j} \in X_\xi \quad \text{in } V[F].$$

Since $q \leq q_m$ and $\pi_m j \in D_q$, this supplies the required witness in $A[D_q] = \{A_j : j \in D_q\}$.

The condition q was chosen before ξ . Although t, d, j were chosen in the argument after F , each is an individual ground object. The dense set in Equation (3.7) therefore belongs to V , and F meets it. No sequence of witnesses for all ξ is being selected or imported into W . For each $\xi < \eta$ the argument shows that q forces $X_\xi \cap A[D_q] \neq \emptyset$ in $V[F]$. It forces the universal statement as well: a condition forcing a counterexample could be strengthened to decide its index in $\check{\eta}$ to one ground ξ , contradicting the assertion for that index. Since this holds for every F containing d_0 , the product condition (d_0, q) forces the universal meeting property.

A ground enumeration of D_q yields a sequence of its labels with an HS name supported by D_q . Distinctness of labels turns that enumeration into a wellordering of $A[D_q]$ in W . Taking the first element of each X_ξ in this wellordering defines the required choice function by Replacement in

W . The argument works below an extension of every starting condition forcing the subset-family hypothesis, and hence proves choice for all such families in W .

Finally let $\langle Y_\xi : \xi < \eta \rangle$ be any family of nonempty sets in W . For $\eta = 0$ use the empty function. Otherwise Proposition 3.6, applied once to the union of the family, gives a surjection $h : A \times \theta \rightarrow \bigcup_{\xi < \eta} Y_\xi$ in W . For each ξ , take the least $\alpha_\xi < \theta$ whose slice meets Y_ξ , and form

$$B_\xi = \{a \in A : h(a, \alpha_\xi) \in Y_\xi\}.$$

These are a family of nonempty subsets of A in W . The subset case chooses $a_\xi \in B_\xi$; then $h(a_\xi, \alpha_\xi)$ is the desired choice from Y_ξ . $\square$

3.6 Failure of Choice and countable-sequence preservation

Proposition 3.8. *The set A is not wellorderable in W . In particular $W \models \neg AC$.*

Proof. Suppose an HS name is forced to be a function from an ordinal η onto A . Choose a forcing condition (d, p) whose diagram range contains the name's support and $\text{supp}_I(d)$. Fix $j \notin D_p$, which exists since $|D_p|^V = \mu < |I|^V$. Strengthen to a product condition (c, t) deciding a ground $\beta < \eta$ with value A_j , and extend the diagram further to have $j \in D_t$.

Choose $\pi \in \text{Fix}(D_p)$ moving all coordinates of t, c outside D_p to fresh ones. Then $t, \pi t$ are independent over p and have a common extension; $c, \pi c$ are also compatible. The function name and $\check{\beta}$ are fixed. Their common product extension would therefore force that one entry has both values A_j and $A_{\pi j}$. They are distinct because $j \notin D_p$ and π moved it freshly. This is a contradiction. $\square$

Proposition 3.9. *For every ground set X , every function $f : \omega \rightarrow X$ belonging to W belongs to V . Consequently W adds no countable sequences of ground elements.*

Proof. Choose a product HS name τ for f . In $V[F]$, the ordinary $\mathbb{P}$ -truth lemma supplies a condition in H forcing that $\tau[F]$ is a function $\check{\omega} \rightarrow \check{X}$. Extend within H to a diagram p whose range covers a support of τ . For each $n < \omega$, the ordinary forcing theorem supplies an extension of p and some $x \in X$ forcing $\tau[F](n) = \check{x}$. Both τ and the check-name parameters are supported by D_p , so Lemma 3.4 makes p itself force that equality. There is exactly one such x , since p forces functionhood.

The forcing relation is definable in $V[F]$. Replacement there therefore forms the sequence of these unique values, a function $g : \omega \rightarrow X$ in $V[F]$. Then $g = f$ in $V[F][H]$, because $p \in H$. By Lemma 3.1, $\mathbb{R}$ adds no countable sequences in the ground set X , so $g \in V$. Thus $f \in V$. The final assertion follows by bounding the ranks of the entries of any countable sequence of ground elements, as in the proof of Lemma 3.1. $\square$

We have obtained the all-rank presentation and internal ordinal choice needed for the final reduction to PP, while the labels already witness failure of full Choice. What remains is the quotient classification: nonwellorderable images of A must be put in bijection with A by one coherent, hereditarily symmetric graph.

4 Exact fibers of a symbolic quotient

Our next objective is a bijection between A and each of its nonwellorderable quotients. A local count of names is insufficient: assignments made in two diagrams must agree whenever those

diagrams occur in one generic filter. We first classify local elements by the subdiagrams on which they are available. We then construct bijections which respect every restriction and every change of coordinates.

Throughout the quotient construction, the cardinal parameters are those of Section 3, with the monoid T supplied there by Theorem 2.2. The forcing $\mathbb{P}$ remains the ground set of diagrams, including when it is used after forcing with $\mathbb{R}$. We use countable closure of $\mathbb{R}$, regularity of $\kappa = \omega_1$, preservation of the ground cardinals μ, μ^+ , and the ground equality $\mu^{\aleph_0} = \mu$. Preservation of smaller cardinals is unnecessary. In particular, μ always denotes the chosen ground ordinal, not a cardinal recomputed from a new continuum. All ground selections below are choices over specified sets in V .

Fix a hereditarily symmetric name $\dot{E}$ and a product condition (d_0, p) symmetrically forcing that $\dot{E}$ is an equivalence relation on A whose quotient is not wellorderable. Enlarge the diagram so that D_p supports $\dot{E}$ and contains $\text{supp}_I(d_0)$. Choose q with $q_e = p$. Our bijection will have support D_q and will be forced below (d_0, q) . Write $G_q = \text{Fix}(D_q)$. This group fixes q, p, d_0 and $\dot{E}$ literally.

We write F for an $\mathbb{R}$ -generic filter containing d_0 , and work temporarily in $V[F]$. A product name τ determines an ordinary $\mathbb{P}$ -name $\tau[F]$ there. Thus an assertion such as $r \Vdash \tau = \sigma$ in this section means ordinary $\mathbb{P}$ -forcing in $V[F]$, using these sections of the product names. The generic notation abbreviates forcing arguments uniform below d_0 ; no claim that a ground permutation acts on one fixed evaluated universe is used.

4.1 Local names and their cardinal bound

Recall the local name collection from Equation (3.5). A normalized subset name means a subset of the ground set

$$\{(\dot{A}_j, (d, s)) : j \in I, (d, s) \in \mathbb{R} \times \mathbb{P}\}.$$

For a diagram r , $\mathcal{X}(r)$ is the ground set of all such names fixed literally by $\text{Fix}(D_r)$. Each is hereditarily symmetric, since its member names are the individual label names. Normalization by the forcing relation retains any given support. When $r \leq q$, define in $V[F]$

$$\mathcal{D}_F(r) = \{\tau \in \mathcal{X}(r) : r \Vdash \tau \text{ is an } \dot{E}\text{-class}\} / \sim_r, \quad \tau \sim_r \sigma \iff r \Vdash \tau = \sigma.$$

These are quotients of ground sets, so they are sets. We sometimes omit F from the notation. Their elements are symbolic classes of names, not yet the evaluated members of A/E in a product extension. Lemma 3.4 applies to the class and equality tests. Consequently, for $r \leq s \leq q$, retaining the same name defines an injection

$$\iota_{s,r} : \mathcal{D}_F(s) \longrightarrow \mathcal{D}_F(r). \tag{4.1}$$

Indeed, if a stronger diagram forced two local names equal, the decision at s would already be equality. These maps compose literally on representatives. Every $j \in D_r$ has a normalized name, supported by D_r , for its E -class: use the forcing relation to name the labels E -related to $\dot{A}_j$. Below the present hypothesis it is an eligible member of $\mathcal{D}_F(r)$.

Lemma 4.1 (Local quotient bound). *For every F as above and every ground $r \leq q$, $|\mathcal{D}_F(r)| \leq \mu$ in $V[F]$.*

Proof. For an element choose a representative τ and a diagram $t \leq r$ forcing $\tau = [\dot{A}_j]_E$ for some $j \in D_t$. The ordinary forcing theorem permits deciding a representative label; extending the

diagram includes its index. Write $r = t_b$. By Cohen restriction in Lemma 3.4, there is $d \in F$, supported in D_t , witnessing this $\mathbb{P}$ -forcing assertion. Record the triple

$$(b, x, t^{-1}d), \quad t(x) = j,$$

where $t^{-1}d$ is the partial binary function on $T \times \kappa$ obtained by replacing each coordinate $t(y)$ with y . Its domain has size less than κ . The ground set of possible records has size at most μ , since $|T| = \mu$, $\kappa \leq \mu$, and $\mu^{<\kappa} = \mu$. Its ground enumeration remains an enumeration in $V[F]$.

Suppose two chosen representatives τ_1, τ_2 have the same record, witnessed by $t_1, t_2, d_1, d_2, j_1, j_2$. The ground bijection $t_1(y) \mapsto t_2(y)$ agrees with the identity on D_r : in both diagrams $r = (t_i)_b$. It maps points outside D_r to points outside D_r . Extend it to a ground permutation π of I fixing D_r . This is possible since the two ranges have size $\mu < |I|$ and their complements have the same cardinality $|I|$. Then

$$\pi t_1 = t_2, \quad \pi d_1 = d_2, \quad \pi j_1 = j_2, \quad \pi \tau_1 = \tau_1.$$

Product symmetry makes d_2 witness at t_2 equality of both τ_1 and τ_2 with $[\dot{A}_{j_2}]_E$. Since $d_2 \in F$, t_2 forces $\tau_1 = \tau_2$ in $V[F]$. Decision at r gives the same equality there. Thus distinct elements have distinct records. Choosing representatives and records uses Choice in the ordinary extension $V[F]$, not in the symmetric model. $\square$

4.2 Availability and exact multiplicities

For $r \leq q$, write $q = r_a$. For $h \in \mathcal{D}_F(r)$ and $j \in D_r$, respectively, define

$$U_h = \{n \sqsubseteq a : h \in \text{ran}(\iota_{r_n, r})\}, \quad (4.2)$$

$$U_j = \{n \sqsubseteq a : j \in D_{r_n}\}. \quad (4.3)$$

An ideal here is nonempty, downward closed, and closed under finite joins. All joins needed within $\downarrow a$ exist.

Lemma 4.2 (Availability ideals). *Each set in Equations (4.2) and (4.3) is an ideal of $\downarrow a$. Every ideal of $\downarrow a$ in $V[F]$ belongs to V .*

Proof. Both sets contain 1 and are downward closed by inclusion of subdiagrams. Suppose $n, n' \in U_h$ and put $k = n \vee n'$. The ranges of $r_n, r_{n'}$ intersect exactly in D_{r_k} . Choose supported names for their representations of h . They are forced equal by r . Lemma 3.5, with root r_k , supplies a D_{r_k} -supported name equal to them. Normalize it without changing support. Decision at r_k shows it is an E -class there: its negation would persist to the extension r . Thus $k \in U_h$. For labels the identical argument is the literal equality of cone intersections, without any name descent.

For the final assertion, apply Lemma 2.1 in $V[F]$. The ground monoid, order, heights and costs retain their meanings there, so every nonempty ideal has a countable cofinal sequence. It consists of ground elements of T , so countable closure of $\mathbb{R}$ puts it in V . Its downward closure, computed with the unchanged ground order, is the original ideal. The argument applies to every ideal in $V[F]$, not only to ground ideals. $\square$

For a nonempty ideal $U \subseteq \downarrow a$, put

$$B_{r,U}(F) = \{h \in \mathcal{D}_F(r) : U_h = U\}, \quad L_{r,U} = \{j \in D_r : U_j = U\}.$$

The second set is ground data; the first can depend on F . These exact fibers, rather than the whole local sets $\mathcal{D}_F(r)$, are the pieces on which a bijection can respect restriction: a label and its assigned class must remain available at precisely the same subdiagrams.

Lemma 4.3 (Exact fibers). *For every $r \leq q$ and every nonempty ideal $U \subseteq \downarrow a$, where $q = r_a$, both $B_{r,U}(F)$ and $L_{r,U}$ have cardinality μ in $V[F]$. The label fiber already has that cardinality in V .*

Proof. The common root p will separate the copies used in this proof. We first find a class with no representation supported by D_p . Copying it into diagrams whose unwanted intersections are exactly D_p will then rule out both coincidences between copies and availability at forbidden subdiagrams, by support descent.

A class unavailable at the root. Enumerate $\mathcal{X}(p)$ in V as $(\tau_\xi : \xi < \lambda)$. The canonical name for the sequence of its evaluations is hereditarily symmetric with support D_p . Explicitly, its entries are the canonical pairs $(\check{\xi}, \tau_\xi)$, each placed under every product condition. Every member of $\text{Fix}(D_p)$ fixes each entry literally. The range of this sequence is wellorderable inside W : map each value to its least index of occurrence. If it contained every E -class, the same map restricted to A/E would wellorder that quotient. The hypothesis therefore implies, ordinarily below (d_0, p) , that some E -class lies outside the evaluations of all names in $\mathcal{X}(p)$.

Fix F containing d_0 . By the ordinary forcing theorem choose $t \leq p$ and $j \in D_t$ such that t forces the class of A_j outside that entire evaluation list. Choose a witnessing Cohen condition $d \in F$ supported in D_t , using Lemma 3.4. Write $p = t_b$. Necessarily $b \neq 1$, since a class represented by a label in D_p has a name in $\mathcal{X}(p)$.

Independent copies with active Cohen conditions. As $p = r_{ae}$, the probe property of Theorem 2.2 supplies μ elements m with $mb = ae$ and the prescribed ideal U . The diagrams r_m are copies of t over p , with pairwise intersection exactly D_p . Choose the ground permutations taking t to these diagrams and fixing D_p , and transport d and the class name along them. Their nonroot coordinate sets are disjoint. This is the role of the earlier extension $q_e = p$: the availability pattern is prescribed among divisors of a , while the common root lies at ae , strictly beyond a .

There are μ copies whose transported Cohen conditions belong to F . To see this without any μ -closure assertion, partition the ground family into μ ground subfamilies of size μ . In each subfamily the conditions extending at least one transported copy of d form a dense set below $d \restriction D_p$. Indeed, a Cohen condition uses countably many coordinate indices; among the pairwise disjoint nonroot ranges there is one it misses, and on the root the two conditions agree. Their union is a Cohen condition. Genericity meets each of these ground dense sets. A hit in each disjoint subfamily yields μ different hits.

Distinct classes with the prescribed exact ideal. Each hit gives a D_{r_m} -supported class name forced at r_m to lie outside the evaluations of $\mathcal{X}(p)$. The diagram r_m extends p but need not extend q ; we use its class name, not a set $\mathcal{D}_F(r_m)$. As r extends r_m , this name represents a symbolic class in the defined set $\mathcal{D}_F(r)$. Two such classes cannot be equal there: support descent over their common root p would supply a name in $\mathcal{X}(p)$ for their common value, contrary to the outside assertion. If $n \in U$, then $n \sqsubseteq m$, so r_n extends r_m and the class is available at r_n . If $n \sqsubseteq a$ and $n \notin U$, the exact probe intersection gives

$$nT \cap mT = aeT, \quad D_{r_n} \cap D_{r_m} = D_p.$$

Availability at r_n would again descend to a name in $\mathcal{X}(p)$, a contradiction. Its exact availability ideal is therefore U . This gives μ distinct elements of $B_{r,U}(F)$; Lemma 4.1 bounds that fiber from above.

The label fiber. For labels use the probe centers $r(m)$. Their availability ideals are exactly $\downarrow m \cap \downarrow a = U$, and distinct probes have distinct centers. This supplies a ground μ -subset of $L_{r,U}$; the upper bound follows from $|D_r| = \mu$. $\square$

5 Components of subdiagrams and their stabilizers

The exact fibers have equal cardinality, but choosing their bijections independently would not preserve assignments when a diagram is replaced by a subdiagram. We first show that restriction identifies entire exact fibers, not merely subsets of them. Following these canonical identifications groups the fibers into components. Only after forming the components will we choose bijections; their stabilizers determine which symmetries those choices must respect.

5.1 Contractions and canonical identifications

A vertex is a pair $v = (r, U)$, where $r \leq q$, $q = r_a$, and U is a nonempty ideal of $\downarrow a$. Vertices form a ground set by Lemma 4.2. For $n \in U$ define a *contraction*

$$(r, U) \longrightarrow (r_n, U_n), \quad U_n = \{x \sqsubseteq c : nx \in U\}, \quad nc = a. \quad (5.1)$$

Left multiplication preserves conditional joins. In fact cancellation and the common-cone description of a join give $n(x \vee y) = nx \vee ny$ whenever these elements have a common right multiple. It follows that U_n is an ideal. Contractions compose; there is at most one contraction between fixed endpoints, since the center of r_n is $r(n)$ and r is injective. Write $B_v(F)$ and L_v for the two fibers at a vertex.

Lemma 5.1 (Contraction identifications). *For every contraction $v \rightarrow w$, the label fibers satisfy $L_v = L_w$, and inclusion of symbolic elements gives a bijection $B_w(F) \longrightarrow B_v(F)$. These identifications commute with composition.*

Proof. Let $v = (r, U)$, $w = (r_n, U_n)$, and first take a symbolic class available at r_n . Its full availability ideal J at r contains n . For $z \sqsubseteq a$, ideal closure gives

$$z \in J \iff z \vee n \in J.$$

Write $z \vee n = nx$. Availability at $r_{z \vee n}$ is exactly the corresponding availability at the subdiagram $(r_n)_x$ of r_n . Thus the restricted ideal at r_n determines the full ideal at r . If that restricted ideal is U_n , the last test is equivalent to $z \vee n \in U$, and hence to $z \in U$. Conversely, an element of $B_v(F)$ is available at r_n because $n \in U$, and its restricted ideal is U_n . This proves surjectivity; injectivity follows from Equation (4.1). For labels use the same argument with literal membership in ranges. Commutation follows because all the symbolic maps retain the same representative names. $\square$

Let a component be a connected component of the vertex system when the direction of contractions is ignored. The successors of each vertex are directed: contract at the join of two contraction indices. Any two vertices in one component have a common successor. Indeed, this holds along a single edge; along a finite path combine a common successor of its initial part with the final edge, using directedness of successors at the shared vertex. The same induction gives a common successor for any finite set of vertices in a component.

Consequently all finite chains of the identifications in Lemma 5.1 are consistent. One can compare a finite chain at a common successor, where both composites are the same map. In a component C , glue the disjoint union of its $B_v(F)$ by these identifications and call the resulting set $B_C(F)$. Each fiber maps bijectively onto $B_C(F)$. In particular, the gluing creates no new identifications within one fiber. All label fibers in C are the same ground subset of I ; denote it by L_C . Both sets have cardinality μ in $V[F]$.

Choose in V a vertex (r, U) of C and a nondecreasing cofinal sequence $(n_i : i < \omega)$ in U , with $n_0 = 1$. Let v_i be its contraction at n_i . This chain is cofinal in the entire component. To check this, compare any vertex with v_0 at a common successor, which is a contraction of v_0 at some $n \in U$. Choose i with $n \subseteq n_i$. Then v_i is a successor of that common successor and hence of the given vertex.

5.2 Only countably many residual symmetries

The group G_q acts on the ground vertex system. On symbolic fibers it acts by transport between generics:

$$T_\pi^F : \mathcal{D}_F(r) \longrightarrow \mathcal{D}_{\pi F}(\pi r), \quad [\tau]_r \longmapsto [\pi \tau]_{\pi r}. \quad (5.2)$$

The forcing symmetry lemma makes this well defined, and it commutes with all contraction identifications. We use the same notation for transport on the glued fibers, and suppress the superscript when its source generic is specified by the argument.

Let G_C be the setwise stabilizer of C in G_q . Define

$$K_C = \bigcup_{i < \omega} \text{Fix}(D_{r_{n_i}}).$$

Each displayed fixer is a subgroup of G_q , because D_q is contained in every chain range. It fixes a vertex and therefore lies in G_C . The ranges decrease, so the subgroups increase and their union is a subgroup. For $\sigma \in G_C$, the moved chain is also cofinal. For every i there is j with $D_{r_{n_j}} \subseteq D_{\sigma r_{n_i}}$. Hence

$$\sigma \text{Fix}(D_{r_{n_i}}) \sigma^{-1} = \text{Fix}(D_{\sigma r_{n_i}}) \subseteq \text{Fix}(D_{r_{n_j}}).$$

Applying the same reasoning to σ^{-1} proves that K_C is normal in G_C .

Lemma 5.2 (Countable stabilizer quotient). *In the ground model, $|G_C/K_C| \leq \aleph_0$.*

Proof. Choose the countable-base tower of Theorem 2.2 so that its initial submonoid T_0 contains all n_i and all right quotients between them. For $\pi \in G_C$, consider its alignments with contractions of the chosen vertex:

$$\mathcal{A}_\pi = \{(l, m) \in \omega \times U : \pi r_{n_l} = r_m\}.$$

This set is nonempty: the moved chain is cofinal in C , so one of its terms is a successor of v_0 . Every second coordinate m of an alignment divides some n_N , by cofinality of (n_i) in U .

To apply Lemma 2.8, fix $(l, m) \in \mathcal{A}_\pi$ and any N . Choose $l' \geq l$ so that $\pi v_{l'}$ is a successor of v_N . Its diagram is r_k for some $k \in U$ with $n_N \subseteq k$. Let $d \in T_0$ be the right quotient $n_l d = n_{l'}$. Then

$$r_k = \pi r_{n_{l'}} = (\pi r_{n_l})_d = r_{md}.$$

Injectivity of r gives $k = md$. Consequently $md \in U$, $n_N \subseteq md$, and $(l', md) \in \mathcal{A}_\pi$. These are exactly the replacement hypotheses of Lemma 2.8. Its conclusion supplies an alignment (l, m) with $m \in T_0$.

There are countably many resulting alignment pairs $(l, m) \in \omega \times T_0$. Two permutations giving the same pair agree pointwise on $D_{r_{n_l}}$; if they are π and ρ , then $\pi^{-1}\rho \in \text{Fix}(D_{r_{n_l}}) \subseteq K_C$. Enumerate the pairs, and assign to each coset the first pair realized by any of its members. Distinct cosets receive distinct pairs. This proves the bound. $\square$

6 Equivariant assignments and the quotient bijection

For each component C we seek a bijection $f_C(F) : L_C \rightarrow B_C(F)$ satisfying

$$f_C(\sigma F)(\sigma j) = T_\sigma^F(f_C(F)(j)) \quad (\sigma \in G_C). \quad (6.1)$$

This is an identity between computations at different generics, not an action on one fixed extension. We first construct a bijection covariant under K_C : its elements fix a sufficiently late subdiagram, so they fix the tails of suitable name codes. We then handle the remaining countable quotient G_C/K_C . Bounded Cohen-row patterns distinguish its cosets and select a transported bijection with the full covariance in Equation (6.1).

6.1 An assignment invariant under a tail fixer

For the chosen chain of C , let $\mathcal{Q}_C$ be the ground set of sequences

$$c = (\tau_i : i < \omega), \quad \tau_i \in \mathcal{X}(r_{n_i}).$$

Say that c represents $b \in B_C(F)$ if, from some index onward, τ_i represents an element of the exact fiber $B_{v_i}(F)$ and its image in $B_C(F)$ is b . Finite initial segments are ignored. Every b has such a code: choose representatives at all chain vertices in $V[F]$, using their bijections with $B_C(F)$. This is a countable sequence of ground name codes, and hence belongs to V by countable closure of $\mathbb{R}$.

Fix a ground wellordering of $\mathcal{Q}_C$, of order type λ_C . Assign to b its least representing code. Distinct b have different codes, so this defines a wellorder $<_F$ of $B_C(F)$.

Lemma 6.1 (Covariance of the least-code order). *For every $k \in K_C$, transport T_k^F is an order isomorphism from $(B_C(F), <_F)$ to $(B_C(kF), <_{kF})$. The group K_C fixes L_C pointwise.*

Proof. Choose N such that k fixes $D_{r_{n_N}}$ pointwise. For $i \geq N$ it fixes r_{n_i} and every name in $\mathcal{X}(r_{n_i})$ literally. Consequently each particular ground code c represents b at F if and only if the same code represents $T_k^F(b)$ at kF . The forcing symmetry lemma proves this on the tail; the definition ignores the remaining finite prefix. Thus the sets of functioning codes are identical, and their minima under the fixed ground wellorder are identical. This proves the order assertion. Also $L_C \subseteq D_{r_{n_i}}$ for every i , so k fixes its every member. $\square$

Let $\theta_C(F)$ be the order type of this order. It is at most λ_C , since it is the order induced on a subset of the ordered code set. It has cardinality μ in $V[F]$. Preservation of μ and μ^+ implies

$$\mu \leq \theta_C(F) < \mu^+, \quad |\theta_C(F)|^V = \mu.$$

Here the order type is an ordinal, hence a ground ordinal. The two preserved cardinal endpoints justify the last equality even if smaller cardinals have collapsed. In V , choose bijections $e_\theta : \mu \rightarrow \theta$ for the set of ordinals $\theta \leq \lambda_C$ with ground cardinality μ . Also fix a ground bijection $l_C : \mu \rightarrow L_C$. If $b_F : \theta_C(F) \rightarrow B_C(F)$ is the unique order isomorphism, define

$$f_C^0(F) = b_F \circ e_{\theta_C(F)} \circ l_C^{-1}.$$

This is a bijection $L_C \rightarrow B_C(F)$. All choices have been restricted to sets; no simultaneous choice over a proper class of ordinals occurs. Lemma 6.1 gives

$$f_C^0(kF)(j) = T_k^F(f_C^0(F)(j)) \quad (k \in K_C, j \in L_C). \quad (6.2)$$

The ground wellorder of all codes need not be invariant under k . The proof uses eventual fixation of each code's entries instead.

6.2 Detecting and choosing a coset

Put $\Gamma_C = G_C/K_C$, writing cosets as πK_C . It is a countable ground group by Lemma 5.2. For each nonidentity coset δ , choose a ground representative γ_δ and points

$$s_{\delta,i} \in D_{r_{n_i}}, \quad \gamma_\delta(s_{\delta,i}) \neq s_{\delta,i} \quad (i < \omega).$$

Such a point exists for every i , since otherwise $\gamma_\delta \in K_C$. Fix an enumeration of the set of nonidentity cosets. For $\alpha < \kappa$ and $\pi \in G_C$, define the pattern

$$P_F^\alpha(\pi K_C) = \left(\left[\left(A_{\pi(s_{\delta,i})}^F \upharpoonright \alpha : i < \omega \right) \right]_{\text{ev}} : \delta \in \Gamma_C \setminus \{1\} \right), \quad (6.3)$$

where $[\cdot]_{\text{ev}}$ denotes equivalence modulo eventual equality of sequences. We regard each row as its binary characteristic function. The equivalence classes in this formula are taken in the ground set of sequences of subsets of α .

This ground interpretation is legitimate. Every $\alpha < \omega_1$ is countable. The bounded row restrictions and all their countably many coordinates, for all detectors, form a countable sequence of ground bits and indices. Since $\mathbb{R}$ adds no such sequences, these data belong to V . Eventual equality between ground sequences has the same meaning in the extension. For all $\alpha < \kappa$ together the possible patterns form a ground set, on which choose a ground wellorder, separately for each α if desired.

Lemma 6.2 (A covariant coset selector). *Formula (6.3) is independent of the representative of πK_C . For each F there is an $\alpha < \kappa$ at which all distinct cosets have distinct patterns. There is a definable selection $c_C(F) \in \Gamma_C$ satisfying*

$$c_C(\sigma F) = \sigma c_C(F) \quad (\sigma \in G_C). \quad (6.4)$$

Proof. If $k \in K_C$, it fixes all $s_{\delta,i}$ once i is sufficiently large. Replacing π with πk therefore changes each detector sequence in only finitely many places, proving representative independence.

Suppose $\pi K_C \neq \pi' K_C$. Use the detector for $\delta = (\pi^{-1}\pi')K_C$ and write $\pi^{-1}\pi' = \gamma_\delta k$ with $k \in K_C$. Eventually $k(s_{\delta,i}) = s_{\delta,i}$, and hence

$$\pi(s_{\delta,i}) \neq \pi'(s_{\delta,i}).$$

Distinct label indices give distinct Cohen rows. For each of these tail positions choose a bit where the two rows differ. The countably many bit ordinals are bounded below regular κ . Above that bound the two restricted sequences disagree at every position on a tail, so are unequal modulo eventual equality. There are countably many pairs of cosets; a further countable supremum supplies one bound that distinguishes all pairs. These choices take place in $V[F]$. They use regularity of κ , not closure of $\mathbb{P}$.

At the least distinguishing α , choose the coset whose pattern is least in the fixed ground wellorder. If Γ_C is trivial, choose its sole element and use $\alpha = 0$. For $\sigma \in G_C$, label transport gives, for every $s \in I$, $A_{\sigma s}^{\sigma F} = A_s^F$. Thus

$$P_{\sigma F}^\alpha((\sigma\pi)K_C) = P_F^\alpha(\pi K_C). \quad (6.5)$$

The actual ground pattern objects on the two sides are equal. Left multiplication permutes the coset set, so the least distinguishing bound is unchanged and the same pattern object is minimal. This proves Equation (6.4). $\square$

$$\begin{array}{ccc}
L_C & \xrightarrow{f_C^0(\pi^{-1}F)} & B_C(\pi^{-1}F) \\
\pi \upharpoonright L_C \downarrow & & \downarrow T_\pi^{\pi^{-1}F} \\
L_C & \xrightarrow{f_C^{\pi K_C}(F)} & B_C(F)
\end{array}$$

Figure 3: Transporting a preliminary bijection, for $\pi \in G_C$. The square commutes: to assign the bottom label j , first compute at $\pi^{-1}j$ in the generic $\pi^{-1}F$, then transport the result to F . The top and bottom computations use $\pi^{-1}F$ and F , respectively.

For a coset $\gamma = \pi K_C$, define a transported preliminary bijection by

$$f_C^\gamma(F)(j) = T_\pi^{\pi^{-1}F} \left(f_C^0(\pi^{-1}F)(\pi^{-1}j) \right), \quad j \in L_C. \quad (6.6)$$

Equivalently, the square in Figure 3 commutes.

This is independent of π . Replacing π with πk inserts k^{-1} in the source generic, and Equation (6.2) cancels its transport; k fixes every label of L_C . Define $f_C(F) = f_C^{c_C(F)}(F)$. Equations (6.4) and (6.6) now give the required Equation (6.1). This calculation explains why the preliminary computation in Equation (6.6) is made at $\pi^{-1}F$. An arbitrary transport of the computation at F would not give the required identity.

6.3 Transport over all components

Choose in V one representative from each G_q -orbit of components and perform the preceding set-sized choices for these representatives. For $C' = \sigma C$, transport $f_C(\sigma^{-1}F)$ by σ to obtain $f_{C'}(F)$. Two possible choices of σ differ by an element of G_C , so Equation (6.1) proves that this definition is independent of the choice. It yields bijections for every component, covariant under all G_q . Pulling back along the canonical maps $B_v(F) \rightarrow B_C(F)$ gives bijections

$$f_{r,U}(F) : L_{r,U} \longrightarrow B_{r,U}(F). \quad (6.7)$$

They commute with every contraction, and their transports under every $\pi \in G_q$ are the corresponding bijections at πF .

We record the definability needed to convert these computations into one symmetric name. The ground sets of vertices, components, name codes, code wellorders, patterns and auxiliary bijections are all sets. The sections of the product names can be constructed in an $\mathbb{R}$ -extension from its generic filter; the ordinary forcing relation for the ground set poset $\mathbb{P}$ is definable there. Symbolic quotients and their gluing use equivalence relations on sets. All remaining steps use set operations, least elements of prescribed wellorders, and unique order isomorphisms. They therefore define, by an ordinary set-theoretic formula in an $\mathbb{R}$ -extension, the predicate

$$\Phi(F, r, j, \tau) : \quad \tau \in \mathcal{X}(r) \text{ represents } f_{r,U_j}(F)(j) \text{ in } \mathcal{D}_F(r). \quad (6.8)$$

Here U_j is ground data from Equation (4.3). One formal implementation uses a ground set of all normalized name codes and its ordinary name-evaluation relation to construct their $\mathbb{P}$ -sections. The variable for a ground code and the actual forcing name that it codes are kept distinct. In particular, we never use the false assertion that an automorphism moves a check-name for a ground code to the check-name for its transported code.

The proved semantic covariance is the identity

$$\Phi(F, r, j, \tau) \iff \Phi(\pi F, \pi r, \pi j, \pi \tau) \quad (\pi \in G_q).$$

These computations are ordinary forcing arguments for this fixed formula. More explicitly, let

$$\dot{F}^\pi = \{(\check{\pi}d, d) : d \in \mathbb{R}\},$$

whose value at F is πF . The preceding covariance proof gives the equivalence, forced by d_0 in $\mathbb{R}$, between $\Phi(\dot{F}, r, j, \tau)$ and $\Phi(\dot{F}^\pi, \pi r, \pi j, \pi \tau)$, with all auxiliary ground choices fixed. Under the name action, $\pi \dot{F}^\pi = \dot{F}$ literally, while check names for the displayed ground arguments remain fixed. Applying ordinary forcing symmetry therefore gives

$$d \Vdash_{\mathbb{R}} \Phi(\dot{F}, r, j, \tau) \iff \pi d \Vdash_{\mathbb{R}} \Phi(\dot{F}, \pi r, \pi j, \pi \tau) \quad (d \leq d_0). \quad (6.9)$$

All auxiliary ground choices are already fixed in the defining formula; covariance was proved for this one construction. The $\mathbb{R}$ -forcing values in this display are ground sets of conditions. Equivalently, if F' contains πd , put $F = \pi^{-1}F'$ and apply covariance to the computation at F ; the converse uses π^{-1} . This generic description abbreviates the fixed-formula argument above and requires no transitive-ground assumption in the consistency deduction.

6.4 One hereditarily symmetric graph

Theorem 6.3 (Quotient classification). *In W , every nonwellorderable surjective image of A is in bijection with A .*

Proof. We first finish the construction for the equivalence relation fixed at the start of Section 4. Use canonical equivariant ordered-pair names and form the product name

$$\dot{J} = \left\{ \left(\langle \dot{A}_j, \tau \rangle^\bullet, (d, r) \right) : \begin{array}{l} r \leq q, j \in D_r, \tau \in \mathcal{X}(r), d \leq d_0, \\ d \Vdash_{\mathbb{R}} \Phi(\dot{F}, r, j, \tau) \end{array} \right\}. \quad (6.10)$$

The superscript denotes the name for the ordered pair. This is a set name: every index ranges over a ground set and the defining forcing relation is a ground relation. Equation (6.9) sends each entry under $\pi \in G_q$ to another entry. Applying π^{-1} gives equality of the sets of entries, so G_q fixes $\dot{J}$ literally. Each member name is a pair of HS names; thus $\dot{J}$ is HS with support D_q . The individual member names may have different supports. Hereditary symmetry does not require all of them to use D_q .

Take a product generic (F, H) containing (d_0, q) . For every $r \in H$ below q , the graph contains exactly the local assignments of Equation (6.7). In detail, whenever $\Phi(F, r, j, \tau)$ holds, the $\mathbb{R}$ truth lemma gives a condition of F forcing it; strengthening with d_0 retains a condition of F , and Equation (6.10) includes its entry. Conversely, every active entry forces its asserted assignment. All classes are represented by names in their symbolic equivalence classes, so the local assignments are complete.

We check the comparison when the diagram is strengthened. Suppose $t \leq r \leq q$, $r = t_b$, and $j \in D_r$. The availability ideal of j at t contains b . Hence

$$(t, U_j \text{ computed at } t) \longrightarrow (r, U_j \text{ computed at } r)$$

is a contraction. Equation (6.7) therefore gives the same assigned symbolic element under inclusion. In particular, its product-name evaluations are equal in every $\mathbb{P}$ -generic containing t .

Any two active entries of $\dot{J}$ occur together below a common stronger diagram in H . At that diagram the union of the exact-fiber bijections is a bijection $D_t \rightarrow \mathcal{D}_F(t)$: availability ideals partition both sets. The preceding comparison proves single-valuedness and injectivity of $\dot{J}^{F,H}$. Distinct symbolic elements are forced unequal there by local decision, and distinct indices have distinct Cohen labels. Thus these are also single-valuedness and injectivity for the actual sets of the graph.

Every label index belongs to a diagram in H below q , by the density of extending diagrams to contain prescribed supports. Therefore the graph has domain A . To reach an arbitrary E -class, take any label belonging to that class and include its index in such a diagram r . Its canonical supported class name represents an element of $\mathcal{D}_F(r)$. The local fiber bijection is onto the fiber containing that element, so the graph reaches the class. This is a pointwise existence argument; no simultaneous selection of representatives of all classes has been made. The range is exactly A/E . The graph belongs to W because its name is HS.

The construction was made on an extension of an arbitrary symmetric condition asserting the nonwellorderable-quotient hypothesis. Applying the symmetric forcing theorem and truth lemma therefore proves in W that every such quotient A/E is in bijection with A . Finally, for a surjection $h : A \rightarrow Y$ in W , form its kernel relation E in W . The map $[a]_E \mapsto h(a)$ is a bijection $A/E \rightarrow Y$, defined by its unique values without choosing class representatives. If Y is nonwellorderable, so is A/E , and composing this bijection with the one just constructed proves the theorem. $\square$

7 From quotient classification to the Partition Principle

We now pass from the classification of quotients of the label set to arbitrary surjections. The reduction uses only three statements inside the symmetric model: every nonempty set is an image of the label set times an ordinal, choice holds for ordinal-indexed families, and every nonwellorderable image of the label set is bijective to it. We first prove this reduction in ZF, independently of the construction.

The ordinal slicing and assembly of injections follow the local-to-global method in Ryan-Smith's analysis of PP [15, Propositions 3.17 and 3.20]. We use the stronger classification of nonwellorderable images stated below and give the complete reduction under these hypotheses.

Lemma 7.1 (The all-set reduction). *Work in ZF. Suppose that a nonempty set A has the following properties.*

1. *For every nonempty set X , there are an ordinal η and a surjection $A \times \eta \rightarrow X$.*
2. *Every ordinal-indexed family of nonempty sets has a choice function.*
3. *Every nonwellorderable surjective image of A is in bijection with A .*

Then PP holds: for all sets X, Y , every surjection $f : X \rightarrow Y$ has an injection $j : Y \rightarrow X$.

Proof. We will use two elementary observations about images of A . If $h : A \rightarrow B$ is onto and $\emptyset \neq S \subseteq B$, then S is also an image of A . Indeed, fix one $s_0 \in S$ and define

$$h_S(a) = \begin{cases} h(a), & h(a) \in S, \\ s_0, & h(a) \notin S. \end{cases}$$

This proves an existence assertion for one given nonempty subset; it does not choose a default element simultaneously for a family. Also, the image of a wellorderable set under any function is

wellorderable: order its image by the least preimages in a fixed wellordering of its domain. Both observations are theorems of ZF.

Fix a surjection $f : X \rightarrow Y$. If $Y = \emptyset$, the empty function is the desired injection. Otherwise X is nonempty. Choose a surjection $s : A \times \eta \rightarrow X$ from the first assumption. Define, for $i < \eta$,

$$\begin{aligned} B_i &= s[A \times \{i\}], \\ Y_i &= f[B_i] \setminus \bigcup_{k < i} f[B_k], \\ S_i &= B_i \cap f^{-1}[Y_i]. \end{aligned} \tag{7.1}$$

Separation and Replacement form these sequences. Every $y \in Y$ belongs to $f[B_i]$ for some i , and its least such index is the unique index for which $y \in Y_i$. Thus the Y_i partition Y . The S_i are pairwise disjoint because their f -images lie in the pairwise disjoint Y_i . Moreover,

$$f \upharpoonright S_i : S_i \rightarrow Y_i$$

is onto for every i . The sets S_i need not cover X .

Before choosing any injections, define

$$\begin{aligned} \mathcal{J}_i &= \{j \in \mathcal{P}(Y \times X) : j \text{ is an injective function,} \\ &\quad \text{dom}(j) = Y_i, \quad \text{ran}(j) \subseteq S_i\}. \end{aligned} \tag{7.2}$$

Power Set and Separation give each candidate set, and Replacement gives the sequence $\langle \mathcal{J}_i : i < \eta \rangle$. We show that every candidate set is nonempty, fixing an arbitrary $i < \eta$.

If Y_i is empty, its empty injection belongs to $\mathcal{J}_i$. Suppose next that Y_i is nonempty and wellorderable. Choose a wellordering of this one set. The nonempty fibers

$$S_i \cap f^{-1}[\{y\}] \quad (y \in Y_i)$$

can be indexed by an ordinal using that wellordering. The second assumption gives a choice function on them. Since different fibers are disjoint, this function is an injection $Y_i \rightarrow S_i$. It therefore belongs to $\mathcal{J}_i$.

Finally, suppose that Y_i is nonwellorderable. Then S_i is nonempty and nonwellorderable, by the second elementary observation and the surjectivity of $f|_{S_i}$. The map $a \mapsto s(a, i)$ surjects from A onto B_i . The first observation makes the nonempty subset $S_i \subseteq B_i$ an image of A , and composition with $f|_{S_i}$ makes Y_i an image of A as well. By the third assumption both S_i and Y_i are bijective to A . Composing one of these bijections with the inverse of the other gives an injection $Y_i \rightarrow S_i$, again in $\mathcal{J}_i$.

We have proved $(\forall i < \eta) \mathcal{J}_i \neq \emptyset$. The pointwise proof selected neither a sequence of wellorderings nor a sequence of quotient bijections. Apply the second assumption now to the already defined sequence of candidate sets. It supplies $\langle j_i : i < \eta \rangle$ with $j_i \in \mathcal{J}_i$. Set

$$j = \bigcup_{i < \eta} j_i.$$

The disjointness of the Y_i makes this a function with domain Y . If $j(y) = j(y')$, that value lies in both corresponding source pieces S_i . Their disjointness makes the indices equal, and injectivity of that j_i gives $y = y'$. Thus $j : Y \rightarrow X$ is an injection.

Every construction in this proof takes place in the same model of ZF. No rank bound on X or Y , and no uniform choice of auxiliary presentations for the slices, has been used. $\square$

Remark 7.2. The third hypothesis of Lemma 7.1 can equivalently be stated for quotients A/E by equivalence relations on A . For a surjection $h : A \rightarrow B$, define

$$a E_h a' \iff h(a) = h(a').$$

The map $[a]_{E_h} \mapsto h(a)$ is a bijection $A/E_h \rightarrow B$. Its graph is defined by the unique common value on each class; it does not require choosing class representatives.

The first hypothesis is the surjective assertion $\text{SVC}(A)$. Its injective counterpart $\text{SVC}^+(A)$ would assert an injection $X \rightarrow A \times \eta$ for every X and some ordinal η . The proof does not assume that counterpart. It follows afterward by applying PP to each presentation surjection. The quotient assertion $\text{PP}(A)$ says that each surjective image of A injects into A . The subset-restricted assertion $\text{PP} \upharpoonright A$ concerns surjections between two arbitrary subsets of A . These are different local statements. The reduction above uses the stronger classification of every nonwellorderable image of A , together with ordinal-indexed choice; it does not replace that classification by quotient injections alone.

The injection obtained in Lemma 7.1 need not respect the fibers of the original surjection. In its wellorderable target case the local injection is a section, but in its nonwellorderable case the two bijections with A give no such requirement. This is precisely the freedom allowed by PP.

7.1 Further cardinal-comparison consequences

The same model separates Choice from two weaker cardinal-comparison principles. Their consequences follow from PP alone and do not require any further forcing construction.

Remark 7.3. For sets X, Y , write $X \leq^* Y$ if $X = \emptyset$ or there is a surjection $Y \twoheadrightarrow X$, and write $X < Y$ if there is an injection $X \hookrightarrow Y$ but no bijection $X \rightarrow Y$. The dual Cantor–Schröder–Bernstein principle CSB^* asserts that $X \leq^* Y$ and $Y \leq^* X$ imply a bijection $X \rightarrow Y$; the Weak Partition Principle WPP asserts that $X \leq^* Y$ implies $\neg(Y < X)$. Both statements quantify over all sets. If either set in the dual-CSB hypothesis is empty, so is the other.

In ZF one has

$$\text{PP} \implies \text{CSB}^* \implies \text{WPP}.$$

Indeed, the empty case of $X \leq^* Y \Rightarrow X \hookrightarrow Y$ has the empty injection, so the first implication follows by applying PP twice and ordinary Cantor–Schröder–Bernstein. For the second, if $X \leq^* Y$ and $Y < X$, the injection $Y \hookrightarrow X$ gives $Y \leq^* X$: for nonempty Y , invert it on its image and extend by one fixed element of Y ; for empty Y , use the definition. Then CSB^* gives a bijection, contrary to $Y < X$. See [3, Definitions 2.1–2.2, Proposition 2.3, and the proofs of Corollary 8.2 and Proposition 9.8].

These ZF implications hold internally in each model. Thus Theorem 1.1 also gives

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \text{CSB}^* + \text{WPP} + \neg \text{AC}).$$

In particular, assuming $\text{Con}(\text{ZF})$, neither $\text{CSB}^* \Rightarrow \text{AC}$ nor $\text{WPP} \Rightarrow \text{AC}$ is provable in ZF. In the separate transitive setting of Theorem 1.2, the same W also satisfies CSB^* and WPP. The assertion AC_{WO} remains true there but is not needed for the implication chain.

7.2 An explicit family without a choice function

The nonwellorderability of the label set already witnesses failure of AC. The following elementary argument also identifies a family of nonempty sets with no choice function.

Lemma 7.4. *In ZF, if a set A is not wellorderable, then $\mathcal{P}(A) \setminus \{\emptyset\}$ has no choice function.*

Proof. Suppose c were a choice function for this family, and let λ be the Hartogs ordinal of A , so there is no injection $\lambda \rightarrow A$. Foundation gives $A \notin A$. By transfinite recursion define $a : \lambda \rightarrow A \cup \{A\}$ as follows. At stage α , put $S_\alpha = A \setminus \text{ran}(a|_\alpha)$ and set

$$a(\alpha) = \begin{cases} c(S_\alpha), & S_\alpha \neq \emptyset, \\ A, & S_\alpha = \emptyset. \end{cases}$$

If every S_α were nonempty, the recursion would choose a different member of A at every stage, giving an injection $\lambda \rightarrow A$. Hence there is a least $\alpha < \lambda$ with $S_\alpha = \emptyset$. Before that stage no value is the sentinel A , and $a|_\alpha$ is a bijection onto A . This wellorders A , a contradiction. Hartogs' theorem and the recursion used here are available in ZF. $\square$

7.3 The resulting symmetric model

We return to the construction. The cardinal choices in the next statement are computed in the ground model. They allow the same argument to be used over a given choice ground as well as over a constructible ground in the consistency deduction.

Over a general ground, the statement is read in its internal forcing interpretation. Ground objects are identified with their canonical check-name images, and ordinals are those of this interpretation. In the sequence assertion, ω is the ground's internal natural-number object, and $z \in V$ means that z is represented by a ground check name. When V is externally transitive, these are the usual evaluated objects and literal inclusions. The two interpretations are distinguished in Theorem 3.3 and constructed separately in Section 8.

Theorem 7.5 (Construction summary). *Let V be a ground model of ZFC, put*

$$\kappa = \omega_1^V, \quad \mu = ((2^{\aleph_0})^+)^V, \quad |I|^V = \mu^+,$$

and use the monoid of Theorem 2.2 and the symmetric system constructed above. In an ordinary generic extension for its product forcing, the resulting symmetric model W satisfies

$$W \models \text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \neg \text{AC}.$$

It contains V and has the same ordinals as V . It adds no countable sequences of ground elements: if $b \in V$ and $z \in W$ is a function $\omega \rightarrow b$, then $z \in V$. The set of labels A is not wellorderable in W , and

$$\mathcal{P}^W(A) \setminus \{\emptyset\}$$

is an explicit family in W with no choice function in W .

Proof. The symmetric system is for one set forcing, with the set group and normal filter specified above. The set-forcing symmetric-model theorem, in its internal form, therefore gives full ZF, the canonical ground embedding, and preservation of the interpreted ordinals. For an externally transitive ground its evaluated hereditarily symmetric interpretation is transitive. These are pure-set models, so Foundation and the full Separation and Replacement schemata are included.

Proposition 3.6 gives the first hypothesis of Lemma 7.1 inside W , and Proposition 3.7 gives the second, for all ordinal-indexed families in W . Theorem 6.3, together with Remark 7.2, gives the third. The lemma proves PP for every surjection in W , with its witnessing injection in W .

Proposition 3.8 proves that A is not wellorderable in W . Applying Lemma 7.4 inside W gives the displayed family and failure of AC there. Finally, Proposition 3.9 supplies the assertion about countable sequences. This last assertion concerns the symmetric model itself; it does not require countable closure of the full product forcing. $\square$

The statement of Theorem 7.5 concerns the model produced by the forcing construction. Extracting an ordinary relative-consistency implication, and obtaining an externally transitive model under the stronger model-existence hypothesis, are separate deductions given next.

8 The consistency deduction

Theorem 7.5 has two uses. For ordinary relative consistency, we must obtain a choice ground from $\text{Con}(\text{ZF})$ and interpret the construction even when that ground is externally ill founded. For the separate transitive-model result, we retain the given transitive choice ground and evaluate names in the usual way. We begin with the precise forcing assertion that both deductions use.

8.1 The internal output of the construction

Write $\mathcal{S} = (\mathbb{Q}, \Gamma, \mathcal{F})$ for a set symmetric system: $\mathbb{Q}$ is a nonempty set poset, Γ is a set group of automorphisms of $\mathbb{Q}$, and $\mathcal{F}$ is a normal filter of subgroups of Γ . Let $\text{HS}_{\mathcal{S}}$ be its class of hereditarily symmetric names. We use a largest condition 1. If necessary, adjoin a fresh largest condition fixed by every automorphism; the original poset remains dense.

The preceding construction is uniform over choice grounds. In particular, Theorem 7.5 gives the following internal assertion in $\text{ZFC} + V = L$: there is a nonempty set symmetric system $\mathcal{S}$ such that

$$1 \Vdash_{\text{HS}_{\mathcal{S}}} \text{PP} \wedge \neg \text{AC} \wedge \text{AC}_{\text{WO}}. \quad (8.1)$$

Here the universal quantifiers of the three sentences range over all internal HS names, with no name-rank cutoff. The standard symmetric-model theorem supplies full ZF for this interpretation; its formula-by-formula justification is given below.

The internal character of this particular construction matters. The monoid is built by recursion along a set ordinal, and its countable-base rearrangement uses an elementary submodel of a set structure H_{Θ} . Its normal forms, diagrams, permutations and supports are sets. In the quotient argument, Equation (3.5) bounds the local name collections by ground sets. The assignment predicate Φ in Equation (6.8) uses those sets, fixed ground choices, and the ordinary forcing relation for the ground poset $\mathbb{P}$ in an $\mathbb{R}$ -extension. Equation (6.10) then forms an actual ground HS graph name. Thus the generic notation in that argument expresses fixed-formula forcing assertions; it is not a construction available only over externally transitive models.

Propositions 3.6 and 3.7 supply the all-rank presentation and ordinal-indexed choice. Together with the quotient graph, the internal ZF reduction in Lemma 7.1 gives PP . Proposition 3.8 supplies $\neg \text{AC}$. The quotient witnesses are produced below a strengthening of every condition asserting the relevant hypothesis. This is the local-witness form of existential forcing, made explicit in Equation (8.2) below; no single symmetric witness is required to work below every condition.

It remains to transfer this forcing assertion to a model obtained from ordinary consistency. We first obtain a choice ground by the constructible interpretation, then define the symmetric extension without external recursion on an ill-founded membership relation.

8.2 The constructible choice ground

Lemma 8.1. *For every axiom α of $\text{ZFC} + V = L$, its relativization α^L is provable in ZF . Consequently*

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZFC} + V = L).$$

Moreover, $\text{ZFC} + V = L$ proves the generalized continuum hypothesis.

Proof. The constructible-universe theorem proves, in ZF , the relativization to L of every ZF axiom, of Choice, and of $V = L$. These are assertions for each individual axiom instance, including the instances of Separation and Replacement. See [17, Theorem 4.7, Lemma 4.10, Corollary 4.13, and Theorems 4.14 and 4.18].

Relativization commutes with Boolean connectives and preserves first-order inference when quantifiers are restricted to the nonempty interpreted domain. A finite contradiction proof from $\text{ZFC} + V = L$ would therefore yield, by relativizing each line and using the finitely many required interpreted axioms, a contradiction proof from ZF . This proves the consistency implication. The same constructible-universe theorem proves GCH in L , hence proves GCH from $\text{ZFC} + V = L$. $\square$

In this choice ground the parameters of the construction are

$$\kappa = \aleph_1, \quad \mu = \aleph_2, \quad I = \aleph_3.$$

These are a convenient specialization of the arbitrary-ground parameters in Section 3. GCH gives

$$\mu^{\aleph_0} = (2^\kappa)^{\aleph_0} = \mu.$$

Since $\kappa = \aleph_1$, this is the bound $\mu^{<\kappa} = \mu$ used in the construction. CH also gives $\kappa^{<\kappa} = \kappa$ here, but that additional equality is not an assumption of the arbitrary-ground argument. The required general bounds were proved in Equation (3.1) and its preceding paragraph. The set constructions and ground wellorderings used earlier are available in $\text{ZFC} + V = L$. In particular, Θ in the monoid rearrangement need only be a sufficiently large regular cardinal; no inaccessible cardinal or satisfaction predicate for the universe is required.

8.3 Symmetric forcing as a first-order interpretation

We now justify the interpretation machinery used with Equation (8.1). The point is not only to obtain the elementary set axioms: every instance of Separation and Replacement must hold for arbitrary-rank names.

Lemma 8.2 (The symmetric forcing schema). *For each fixed formula φ in the language of set theory, the relation*

$$p \Vdash_{\text{HS}_S} \varphi(\tau_1, \dots, \tau_n)$$

is definable in ZFC , with S as a parameter and $\tau_1, \dots, \tau_n \in \text{HS}_S$. It satisfies the forcing rules for first-order logic with equality and the symmetry lemma. For each axiom α of ZF , including each Separation and Replacement instance, ZFC proves

$$S \text{ is a set symmetric system} \implies 1 \Vdash_{\text{HS}_S} \alpha.$$

Proof. Atomic forcing is defined by the usual recursion on names. For a fixed formula, extend these atomic definitions through its finitely many connectives and quantifiers, restricting quantified names to HS_S . For example, define

$$D_{\varphi, \vec{\tau}} = \{q \in \mathbb{Q} : (\exists \sigma \in \text{HS}_S) q \Vdash_{\text{HS}_S} \varphi(\sigma, \vec{\tau})\}.$$

Existential forcing is characterized by

$$p \Vdash_{\text{HS}_S} \exists x \varphi(x, \vec{\tau}) \iff D_{\varphi, \vec{\tau}} \text{ is dense below } p. \quad (8.2)$$

The set $D_{\varphi, \vec{\tau}}$ exists by Separation from $\mathbb{Q}$. The class of names and the hereditary symmetry predicate are definable by set-theoretic recursion. Thus all the displayed relations are ordinary first-order predicates. This construction is a schema indexed externally by formulas; it asserts no single satisfaction predicate for all formulas.

The logical and symmetry rules follow by induction on a fixed formula, with the internal name recursion supplying the atomic case. The resulting symmetric-model theorem and its forcing relation are standard; see [10, Equations (15.39)–(15.41) and Lemma 15.51] and [11, Section 2.1, p. 4]. We recall the parts that ensure the full axiom schemata.

Fix an arbitrary formula $\varphi(x, \vec{a})$ and hereditarily symmetric names $\sigma, \vec{\tau}$ for its set and parameters. A Separation name uses the pairs $(\rho, q) \in \text{dom}(\sigma) \times \mathbb{Q}$ for which

$$q \Vdash_{\text{HS}_S} \rho \in \sigma \wedge \varphi(\rho, \vec{\tau}).$$

The intersection of the stabilizers of σ and the finitely many parameter names fixes this name. That intersection belongs to $\mathcal{F}$, and every member name is hereditarily symmetric. The atomic forcing rules and Equation (8.2) verify that it is the required subset. No restriction on φ or on the rank of σ has been imposed.

Write $\text{nrk}(\eta)$ for the name-rank

$$\text{nrk}(\eta) = \sup\{\text{nrk}(\zeta) + 1 : (\zeta, r) \in \eta\}.$$

For Replacement, fix a formula $\psi(x, y, \vec{a})$, a domain name σ , and parameter names $\vec{\tau}$. For each pair $(\rho, q) \in \text{dom}(\sigma) \times \mathbb{Q}$ for which some hereditarily symmetric name η satisfies

$$q \Vdash_{\text{HS}_S} \psi(\rho, \eta, \vec{\tau}),$$

take the least possible name-rank of such an η . This defines an ordinal-valued function on a ground set. Replacement in the ground bounds all these ranks strictly below one ordinal β . All $\mathbb{Q}$ -names of name-rank less than β form a set. Consequently

$$\dot{B}_\beta = \{(\eta, 1) : \eta \in \text{HS}_S, \text{nrk}(\eta) < \beta\}$$

is a set name. Every automorphism preserves name-rank and hereditary symmetry, so $\dot{B}_\beta$ is fixed by the whole group and belongs to HS_S .

Below a condition forcing totality of the relation on σ , local witnesses from Equation (8.2) can therefore be replaced, at the same witnessing conditions, by names occurring in $\dot{B}_\beta$. This gives a set bounding the required witnesses. When the relation is functional, Separation in that set gives its image. The bound β depends on the particular domain name and formula; it is not a bound on the universe of names.

For Power Set, normalize a name for a subset of σ using member names from $\text{dom}(\sigma)$ and the forcing tests for membership in that subset. The normalized names lie in the ground set

$\mathcal{P}(\text{dom}(\sigma) \times \mathbb{Q})$. Use all hereditarily symmetric candidates, under the conditions forcing that they are subsets of σ . The stabilizer of σ permutes this collection, giving a hereditarily symmetric power-set name. Every symmetric subset has such a normalization, so the name supplies the entire internal power set. Check names and the canonical pairing and union constructions supply the remaining elementary existence axioms.

The usual internal name-rank argument verifies Foundation. Equivalently, the interpreted hereditary class is membership-closed inside the ordinary forcing extension: an element of an HS-valued set has an active member name of an HS name. Extensionality and Foundation then inherit their witnesses from the ordinary extension. This argument concerns first-order Foundation and does not assume that an arbitrary model is externally well founded.

All these name constructions and forcing verifications are proofs in the ground theory, for the chosen formula. They prove the asserted schema. They use local existential witnesses rather than a maximum principle for symmetric forcing; that maximum principle need not hold [12, Proposition 10.26]. $\square$

8.4 The extension of an arbitrary countable model

Let $M = (|M|, E^M)$ be a countable, possibly ill-founded model of ZFC. Suppose M contains a set symmetric system $\mathcal{S}$. Its internal poset determines the external set

$$Q^M = \{p \in |M| : M \models p \in \mathbb{Q}\},$$

with the order interpreted by M . An M -generic filter means a filter on this poset meeting the interpretation of every $D \in |M|$ for which M asserts that D is dense in $\mathbb{Q}$.

Such a filter exists through any prescribed condition. Enumerate these dense-set codes externally and choose a descending sequence meeting them successively. Internal density really supplies an external strengthening in Q^M , since its witnesses are elements of the domain of M . The upward closure of the chosen sequence is the required generic filter G . No closure property of $\mathbb{Q}$ and no well-foundedness of E^M are needed here.

Let $\mathcal{N}^M$ be the external set of elements of M which M regards as $\mathbb{Q}$ -names. Define

$$\sigma \sim_G \tau \iff (\exists p \in G) M \models p \Vdash \sigma = \tau, \quad (8.3)$$

$$[\sigma]_G E_G [\tau]_G \iff (\exists p \in G) M \models p \Vdash \sigma \in \tau. \quad (8.4)$$

The equality and substitution laws proved internally in M , together with directedness of G , make $\sim_G$ an equivalence relation and make E_G independent of representatives. The quotient of all internal names is the ordinary extension structure $M[G]$. Its substructure W consists of the equivalence classes having a representative which M regards as hereditarily symmetric. Both are external set structures, since $|M|$ is a set. The ordinary forcing quotient over nontransitive models is described in [18, Section 4, p. 12].

Lemma 8.3 (Truth for the symmetric quotient). *For each standard first-order formula φ and internal hereditarily symmetric names $\vec{\tau}$,*

$$W \models \varphi([\vec{\tau}]_G) \iff (\exists p \in G) M \models p \Vdash_{\text{HS}_S} \varphi(\vec{\tau}).$$

In particular $W \models \text{ZF}$.

Proof. Induct externally on the finite syntax of the standard formula, using atomic formulas, conjunction, negation, and existential quantification. The atomic case is Equations (8.3) and (8.4);

atomic symmetric forcing agrees with ordinary forcing. Conjunction uses a common strengthening of two conditions in G and monotonicity.

For negation, the set of conditions deciding the particular formula is an internally definable dense subset of $\mathbb{Q}$. It belongs to M by the Separation instance for its defining forcing predicate. Genericity gives a deciding condition in G . The induction hypothesis excludes the wrong decision, and directedness excludes conditions in G forcing opposite answers.

Suppose $p \in G$ symmetrically forces an existential statement. The set of local witness conditions in Equation (8.2), together with the conditions incompatible with p , is another internally definable dense set. Genericity meets its first part. The internal witness is an actual element of $|M|$ which M regards as an HS name. Its equivalence class is a witness in W by induction. Conversely, a witness in W has an internal HS representative; apply induction to that representative and the elementary existential forcing rule.

This proves the truth equivalence for every standard formula. Lemma 8.2, applied in M to each standard ZF axiom instance, now gives $W \models \text{ZF}$. The proof never evaluates a name by external recursion over E^M . Nor does it require a truth predicate in M for all formulas, including its possible nonstandard formula codes. $\square$

The membership-closed assertion used above can also be seen directly in this quotient. If $[\sigma]_G E_G [\tau]_G$ and τ is internally HS, the atomic forcing rule and genericity provide an active member name ρ of τ with $[\rho]_G = [\sigma]_G$. Heredity makes ρ internally HS. Thus W and $M[G]$ have the same members of each W -object. This relational form of transitivity does not make E_G an externally well-founded relation.

8.5 Applying the construction

The internal forcing assertion (8.1) can now be used in a countable choice ground whether or not its membership relation is externally well founded.

Proof of Theorem 1.1. Assume $\text{Con}(\text{ZF})$. By Lemma 8.1, $\text{ZFC} + V = L$ is consistent. Completeness and downward Löwenheim–Skolem give a countable model M of this theory. No transitivity or external well-foundedness is asserted.

Interpret the construction inside M and choose an external M -generic filter on its internal poset as above. Form the symmetric quotient W using Equations (8.3) and (8.4). Lemma 8.3 gives full ZF in W . The same lemma, applied to Equation (8.1), gives

$$W \models \text{ZF} + \text{PP} + \neg\text{AC} + \text{AC}_{\text{WO}}.$$

Soundness therefore yields $\text{Con}(\text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \neg\text{AC})$, as required.

Equivalently, one can extract the consistency implication directly from finite proofs. A purported contradiction proof from the target theory uses only finitely many ZF instances. Translate its lines into symmetric forcing. The logical forcing rules preserve its inferences; Lemma 8.2 supplies its ZF axioms; and Equation (8.1) supplies PP, AC_{WO} , and $\neg\text{AC}$. This would make a nonempty poset force a contradiction, contrary to its atomic forcing laws. It therefore gives a contradiction proof in $\text{ZFC} + V = L$, which the constructible interpretation translates into one in ZF. This argument uses no stronger model-existence assumption. $\square$

8.6 The separate transitive-model conclusion

Proof of Theorem 1.2. Work in a ZFC metatheory, and let V be the given countable transitive model of ZFC. Carry out Theorem 7.5 inside V , using its parameters

$$\kappa = \omega_1^V, \quad \mu = ((2^{\aleph_0})^+)^V, \quad I = (\mu^+)^V.$$

These are the arbitrary-choice-ground parameters of the construction.

Enumerate externally the dense subsets of the constructed poset which belong to V , and recursively choose a descending sequence meeting them. Its upward-generated filter G is V -generic. Here genuine well-founded recursion on ground names is available, since V is transitive. Define

$$W = \{\tau^G : \tau \in \text{HS}^V\}.$$

The set symmetric-extension theorem gives transitive models $V \subseteq W \subseteq V[G]$ with the same ordinals, and gives all of ZF in W . Theorem 7.5 supplies PP, $\neg\text{AC}$, and AC_{WO} in this same W , together with its stated preservation of countable sequences from any fixed ground set. To obtain the last assertion of Theorem 1.2, let $f \in W$ be a countable sequence of elements of V . Its rank in $V[G]$ is bounded by a ground ordinal α , since forcing preserves the ordinals. Ranks of ground sets are absolute between these transitive models, so the range of f lies in the ground set $(V_\alpha)^V$. The preservation assertion of Theorem 7.5 applied to this set gives $f \in V$. This proves the stated transitive-model conclusion.

This is a separate model-existence deduction. The ordinary consistency proof permits an externally ill-founded ground and does not itself assert the existence of a transitive model. $\square$

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